# Chapter # 06
## Endocrine System of Man

**Student Learning Outcomes**

**After studying this chapter, students will be able to:**
- [B-12-G-23] State the role of hormones as chemical messengers.
- [B-12-G-24] Describe the chemical nature of hormones and correlate it with important hormones.
- [B-12-G-25] Locate the endocrine glands in human body name the hormones they release and their functions; (pituitary, thyroid, parathyroid, pancreas, adrenal, gonads).
- [B-12-G-26] Relate problems associated with the imbalance of these hormones.
- [B-12-G-27] Explain the neurosecretory role of hypothalamus.
- [B-12-G-28] Outline the concept of Feedback mechanism of hormones. Describe positive feedback with reference to oxytocin and negative feedback with reference to insulin and glucagon.

The human body depends on precise regulation of its internal environment to maintain **homeostasis** and support life processes. This regulation occurs through two major systems: the **nervous system,** which uses rapid electrical impulses, and the **endocrine system**, which relies on **chemical coordination**.

**Chemical coordination** is the regulation of body functions through **hormones** rather than nerve impulses. Unlike the nervous system, which can directly control only certain tissues, most body cells need regulation without direct neural contact. To meet this need, the endocrine system secretes hormones into the **bloodstream**, allowing communication with target cells and organs throughout the body. In this way, the endocrine system acts as a powerful and specialized mechanism for long-term regulation.

### Importance of Chemical Coordination

Hormones influence almost every aspect of human physiology. They regulate **metabolic activities**, control the **rate of biochemical reactions**, assist in the **transport of molecules** across cell membranes, and stimulate or inhibit **glandular secretions**. In addition, hormones play a vital role in **growth and development, reproductive functions,** and the maintenance of overall body balance. Without chemical coordination, the internal stability of the body would be disrupted, threatening survival.

### Hormones – Messengers of the Endocrine System

**Hormones** are organic chemical messengers secreted by specific cells, usually located in **endocrine glands.** Once released into the blood, they travel to distant targets where they modify the activity of cells, tissues, or organs. Although often present in very small amounts, hormones have profound effects, either stimulating or suppressing essential biological processes.

To date, scientists have identified more than 50 distinct hormones in the human body. They regulate diverse functions such as metabolism, blood pressure, blood glucose level, body temperature, fluid and electrolyte balance, sleep–wake cycles, and even mood and emotional state.

### Chemical Nature of Hormones

Most hormones fall into well-defined categories based on their chemical structure:
- Amines (derived from amino acids)
- **Peptides and proteins**
- ***Glycoproteins***
- ***Steroids*** (derived from cholesterol)

This chemical diversity allows hormones to act in highly specific ways, ensuring that the body's internal environment remains stable and that organ systems work in coordination.

SLO [B-12-G-23]
**State the role of hormones as chemical messengers.**

## ROLE OF HORMONES AS CHEMICAL MESSENGERS

### Hormones and Chemical Coordination

The human body requires continuous regulation to maintain **homeostasis** and to coordinate the activities of different organs. While the **nervous system** provides rapid control through electrical impulses, another equally important system works more slowly but with longer-lasting effects—the **endocrine system**. It functions through **hormones**, which act as **chemical messengers** to communicate between cells, tissues, and organs.

### Definition and Nature of Hormones

**Hormones** are organic chemical substances secreted in minute quantities by **endocrine glands**. Unlike exocrine glands, which release their secretions into ducts, endocrine glands are **ductless** and pour hormones directly into the **bloodstream**. The blood then transports these hormones to **target organs or tissues**, where they bind to specific **receptors** and regulate cellular activities.

### Role of Hormones as Messengers

The role of hormones is to act as messengers that **transmit information chemically** from one part of the body to another. By doing so, they:

- Control the **rate of metabolic reactions** (e.g., thyroid hormones regulating metabolism).
- Maintain **homeostasis**, such as the regulation of **blood glucose levels** by insulin and glucagon.
- Influence **growth and development**, as seen with growth hormone and sex hormones.
- Regulate **reproduction**, including sexual differentiation, menstrual cycle, and pregnancy.
- Adjust **mood, emotions, and biological rhythms**, such as the sleep–wake cycle influenced by melatonin.

> **SCIENCE TITBITS**
> In 1902, Bayliss and Starling, prepared an extract from the duodenum which stimulated secretion of pancreatic digestive juices when it was injected into the bloodstream. They called the product 'secretin', and coined the term 'hormone', meaning 'to excite' or 'to set in motion.'

### Key Features of Hormonal Communication

Unlike nerve impulses, which are fast and localized, hormonal communication is:

- Slower in action but has more prolonged effects.
- Widespread, as hormones can reach almost all parts of the body through the bloodstream.
- Highly specific, because only cells with the right receptors can respond to a given hormone.

**In summary,** hormones act as **chemical messengers** that enable the body to coordinate complex activities and maintain stability of the internal environment. They ensure that different organ systems function in harmony, supporting survival, growth, reproduction, and overall health.

SLO [B-12-G-24]
**Describe the chemical nature of hormones and correlate it with important hormones.**

## CHEMICAL NATURE OF HORMONES AND THEIR CORRELATION WITH IMPORTANCE

### 1. Chemical Nature of Hormones

**Hormones** are specialized **organic compounds** secreted in minute quantities by endocrine glands. Despite being present *in very small* amounts, they exert powerful effects on specific **target cells. Their chemical nature** determines many key aspects of their function: how they are **transported in the bloodstream,** how they **interact with receptors,** and how quickly or slowly they bring about physiological responses. Broadly, hormones can be classified into the following categories:

**(a) Steroid Hormones**

17

**Steroid hormones** are **lipid-soluble molecules** synthesized from **cholesterol**. Because they are non-polar, they can easily diffuse through **cell membranes** and bind to **intracellular receptors** in the cytoplasm or nucleus. Once bound, the hormone–receptor complex directly influences **gene expression** and protein synthesis, producing long-lasting effects.

- **Source:** Secreted mainly by the **adrenal cortex**, **gonads (testes and ovaries)**, and the **placenta.**
- **Examples: Cortisone** and **aldosterone** (adrenal cortex), which regulate stress responses and salt-water balance; **testosterone** (testes), **estrogen** and **progesterone** (ovaries), which control sexual development, reproduction, and pregnancy.

Because of their action on gene regulation, steroid hormones usually act **more slowly** than peptide hormones, but their effects are **longer lasting**.

**(b) Proteinaceous Hormones (Peptides and Polypeptides)**

**Proteinaceous hormones** include **peptides, polypeptides, and proteins**, all of which are composed of chains of **amino acids**. These hormones are **water-soluble**, meaning they cannot pass directly through the lipid bilayer of cell membranes. Instead, they bind to **receptors on the cell surface** and activate **secondary messenger systems** (such as cyclic AMP), leading to rapid changes in cell activity.

- **Source:** Produced by a wide range of endocrine glands, especially the **anterior pituitary** and the **pancreas.**
- **Examples: Growth hormone (GH)** stimulates body growth; **thyroid-stimulating hormone (TSH)** regulates thyroid gland activity; **follicle-stimulating hormone (FSH)** and **luteinizing hormone (LH)** regulate reproductive cycles; **insulin** (pancreas) lowers blood glucose by promoting glucose uptake in cells.

These hormones generally act **faster** than steroid hormones, but their effects are often **short-lived.**

**(c) Catecholamines**

**Catecholamines** are small, water-soluble hormones derived from the amino acid **tyrosine**. They are secreted by the **adrenal medulla** and play a crucial role in the **"fight or flight" response** during stressful situations. Because they are water-soluble, catecholamines bind to **cell surface receptors** and trigger rapid metabolic changes.

- **Examples: Adrenaline (epinephrine)** increases heart rate, blood pressure, and glucose availability, preparing the body for emergency action. **Noradrenaline (norepinephrine)** complements these effects by constricting blood vessels and maintaining blood pressure.

Catecholamines act very **quickly**, but their effects are relatively **short in duration**, making them ideal for immediate stress responses.

**(d) Amino Acid Derivatives**

Some hormones are simple **derivatives of single amino acids**, such as tyrosine. These hormones may behave like peptides (water-soluble, acting on surface receptors) or like steroids (lipid-soluble, acting inside cells), depending on their structure.

- **Source:** Secreted mainly by the **thyroid gland.**
- **Example: Thyroxine (T4)**, which regulates **metabolic rate, growth, and energy balance**. Because of its lipid-like nature, thyroxine can enter cells and influence **long-term metabolism**.

**(e) Other Peptide Hormones**

This group includes hormones made of **short or intermediate chains of amino acids**. They show great diversity in function and are secreted by the **pituitary gland** and other endocrine organs.

- **Examples: Melanocyte-stimulating hormone (MSH)** controls pigmentation; **oxytocin** stimulates uterine contractions and milk ejection; **vasopressin (antidiuretic hormone, ADH)** regulates water balance; **adrenocorticotrophic hormone (ACTH)** stimulates adrenal cortex activity; **calcitonin** lowers blood calcium levels; **parathormone (PTH)** raises blood calcium levels.

These hormones act rapidly and play vital roles in maintaining **homeostasis**, reproduction, and stress adaptation.

## 2. Correlation Between Chemical Nature and Importance

The **chemical nature** of a hormone largely determines **how it acts in the body** and the specific **physiological processes** it controls. Understanding this correlation helps explain why different hormones produce rapid or long-lasting effects, act locally or throughout the body, and regulate particular biological functions. 17 17

### (a) Steroid Hormones and Long-Term Regulation

**Steroid hormones** are **lipid-soluble**, allowing them to **diffuse across cell membranes** and bind to **intracellular receptors** in the cytoplasm or nucleus. Once bound, they influence **gene expression**, resulting in the synthesis of specific proteins that modify cellular function. Because of this mechanism, steroid hormones generally act **slowly** but produce **long-lasting effects**.

- **Physiological importance:** They are critical for **stress regulation** (cortisone), **salt and water balance** (aldosterone), and **reproductive functions** (testosterone, estrogen, progesterone). These hormones coordinate **metabolism, reproductive development, and adaptation to stress**, making them essential for long-term homeostasis.

### (b) Proteinaceous Hormones and Rapid Responses

**Proteinaceous hormones**, including **peptides and polypeptides**, are **water-soluble** and cannot pass through the lipid membrane of target cells. Instead, they act on **surface receptors**, activating **secondary messenger systems** such as cyclic AMP or calcium ions. This mechanism enables **rapid responses** to physiological changes.

- **Physiological importance:** Proteinaceous hormones regulate **growth** (growth hormone, GH), **blood glucose levels** (insulin), and **reproductive cycles** (FSH and LH). Their fast action is crucial for processes that require immediate adjustment, such as blood sugar control after meals or rapid hormonal feedback in reproduction.

### (c) Catecholamines and Emergency Action

**Catecholamines** are small, **water-soluble hormones** derived from **tyrosine**, secreted by the **adrenal medulla** in response to stress. They act **almost instantly** on cell surface receptors, triggering metabolic and cardiovascular changes.

- **Physiological importance: Adrenaline (epinephrine)** and **noradrenaline (norepinephrine)** initiate the **"fight or flight" response**, increasing heart rate, blood pressure, and energy supply. Their rapid action prepares the body to cope with **emergency situations**, demonstrating the importance of water-soluble hormones for immediate physiological adjustments.

### (d) Amino Acid Derivatives and Metabolic Control

Some hormones, such as **thyroxine**, are **amino acid derivatives** that exhibit both peptide- and steroid-like properties. They often enter target cells and influence **long-term metabolic activities**.

- ***Physiological* importance: Thyroxine,** secreted by the thyroid gland, regulates **metabolism, growth, and energy balance**. Its slow but prolonged action ensures **consistent metabolic rate**, proper **growth**, and **development**, highlighting the significance of hormones that act over extended periods.

### (e) Other Peptide Hormones and Diverse Functions

A group of **short or intermediate peptide hormones** regulate highly specialized physiological processes.

- **Physiological importance: Oxytocin** stimulates uterine contractions during childbirth and milk ejection during lactation. **Vasopressin (ADH)** maintains **water balance** by controlling kidney function. **Calcitonin** and **parathormone (PTH)** regulate **blood calcium levels**, and **ACTH** stimulates the adrenal cortex to release stress-related steroid hormones. These hormones exemplify how peptides mediate **specific, rapid, and targeted regulatory functions** throughout the body.

Each class of hormone differs in its solubility, mode of transport in the bloodstream, receptor interaction, and mechanism of action. This chemical diversity enables hormones to regulate a broad spectrum of physiological processes throughout the body. In summary, hormones can be classified into **steroids, proteinaceous hormones, catecholamines, amino acid derivatives**, and **other peptide hormones**. Their **chemical structure** determines their **mode of action** and their **physiological importance**, from rapid stress responses to long-term regulation of metabolism, growth, reproduction, and homeostasis.

## HORMONES, THEIR COMPOSITION AND ROLE

**1. Which of the following best defines a hormone?**
A. An enzyme that catalyzes cell reactions
B. A nerve impulse carrier
C. An organic chemical messenger secreted into the blood
D. A neurotransmitter acting at a synapse

**2. What characteristic allows hormones to act even at very low concentrations?**
A. They are stored in large quantities
B. They act only on muscle cells
C. They bind specifically to high-affinity receptors
D. They are insoluble in blood

**3. Hormones reach their target cells through:**
A. Lymph B. Direct diffusion
C. Nerve fibers D. Bloodstream

**4. Which gland secretes thyroid-stimulating hormone (TSH)?**
A. Thyroid gland B. Adrenal cortex
C. Pancreas D. Pituitary gland

**5. Which of the following is a steroid hormone?**
A. Insulin B. Adrenaline
C. Cortisone D. Oxytocin

**6. What is the source of adrenaline and noradrenaline?**
A. Adrenal cortex B. Thyroid gland
C. Adrenal medulla D. Pancreas

**7. Thyroxine is chemically classified as a:**
A. Peptide B. Protein
C. Amino acid derivative D. Steroid

**8. Hormones affect only specific target cells because:**
A. They are universal messengers
B. Target cells have specific receptors
C. All cells respond to all hormones
D. They are distributed through lymph

**9. Which of the following hormones is secreted by the pancreas?**
A. Vasopressin B. Insulin
C. Thyroxine D. Aldosterone

**10. Which of the following is a peptide hormone?**
A. Estrogen B. Testosterone
C. Oxytocin D. Cortisone

**11. A hormone secreted by the thyroid gland is:**
A. Insulin B. Thyroxine
C. TSH D. Cortisol

**12. What is the function of TSH in the body?**
A. Stimulates growth of bone
B. Stimulates adrenal medulla
C. Stimulates the thyroid gland to secrete hormones
D. Regulates blood glucose levels

**13. Which hormone controls water balance in the body?**
A. Vasopressin B. Insulin
C. Adrenaline D. Progesterone

**14. Estrogen and progesterone are produced by:**
A. Adrenal cortex B. Ovaries
C. Anterior pituitary D. Thyroid gland

**15. The "lock and key" model is used to describe:**
A. Hormonal degradation
B. Hormone-receptor specificity
C. Hormone secretion
D. Neural transmission

**16. Which of the following is NOT a proteinaceous hormone?**
A. FSH B. Insulin
C. Testosterone D. TSH

**17. Which hormone is responsible for lowering blood glucose?**
A. Cortisol B. Glucagon
C. Insulin D. Adrenaline

**18. ACTH is secreted by:**
A. Thyroid gland B. Adrenal medulla
C. Posterior pituitary D. Anterior pituitary

**19. Which of these is a catecholamine?**
A. Aldosterone B. Testosterone
C. Noradrenaline D. Oxytocin

**20. Hormonal communication between two glands, such as the pituitary and thyroid, is an example of:**
A. Autocrine signaling
B. Neural feedback
C. Endocrine-to-endocrine communication
D. Paracrine control

**Answer Key**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. C | 2. C | 3. D | 4. D | 5. C | 6. C | 7. C | 8. B | 9. B | 10. C |
| 11. B | 12. C | 13. A | 14. B | 15. B | 16. C | 17. C | 18. D | 19. C | 20 C |

## Short Questions

**Q.1 What is a hormone?**

**Ans.** A hormone is a **chemical messenger** secreted by endocrine glands directly into the bloodstream. It travels to specific **target tissues or organs** and binds to **specific receptors**, even in minute concentrations, to regulate physiological activities. Hormones control **growth, metabolism, reproduction, and homeostasis**, ensuring that various body systems function in a coordinated manner.

> **MARKING SCHEME (3 Marks)**
> - Definition (2 marks), function (1 mark)

**Q.2 What is meant by chemical coordination?**

**Ans.** Chemical coordination refers to the **regulation and integration of bodily functions through hormones**. Unlike nerve impulses, which are rapid and short-lived, hormonal signals are **slower but longer-lasting.** Hormones travel via the **bloodstream** to reach **distant target organs or tissues**, modulating processes such as metabolism, growth, reproduction, and homeostasis in a coordinated manner across the body.

> **MARKING SCHEME (3 Marks)**
> - Definition (2 marks), role of hormones (1 mark)

**Q.3 Name four important functions regulated by hormones.**

**Ans.** Hormones regulate:

1. **Metabolism** – controlling energy production and utilization.
2. **Growth and development** – influencing body size, tissue differentiation, and maturation.
3. **Fluid and electrolyte balance** – maintaining proper water and salt levels.
4. **Reproductive processes and mood regulation** – controlling fertility, sexual development, and emotional behavior.
5. These functions ensure **homeostasis** and coordinated physiological responses throughout the body.

> **MARKING SCHEME (3 Marks)**
> - 1 mark each for any three main functions

**Q.4 Differentiate between steroid and peptide hormones.**

**Ans.** **Steroid hormones** are **lipid-soluble** molecules derived from **cholesterol**, such as estrogen, testosterone, and cortisol. They can **diffuse through cell membranes** and bind to **intracellular receptors**, influencing gene expression.
**Peptide hormones** are **water-soluble** chains of amino acids, like insulin, glucagon, and TSH. They **cannot cross cell membranes** and bind to **surface receptors**, triggering secondary messenger pathways. Steroid hormones generally have **longer-lasting effects**, while peptide hormones act **rapidly but briefly.**

> **MARKING SCHEME (3 Marks)**
> - Steroid hormones (1.5 marks), peptide hormones (1.5 marks)

**Q.5 Give the source and function of insulin.**

**Ans.** Insulin is a peptide hormone secreted by the beta cells of the pancreas. Its primary function is to lower blood glucose levels by promoting glucose uptake into liver, muscle, and fat cells. It also stimulates glycogen synthesis, inhibits gluconeogenesis, and ensures proper energy storage and utilization in the body.

> **MARKING SCHEME (3 Marks)**
> - Source (1 mark), function (2 marks)

**Q.6 What are catecholamines? Give examples.**

**Ans.** Catecholamines are **hormones derived from the amino acid tyrosine,** secreted mainly by the **adrenal medulla.** They are involved in the **fight-or-flight response,** preparing the body for stress by increasing heart rate, blood pressure, and glucose levels. **Examples include adrenaline (epinephrine) and noradrenaline (norepinephrine).**

> **MARKING SCHEME (3 Marks)**
> - Definition/characteristics (2 marks), examples (1 mark)

**Q.7 Explain hormone action using the lock and key model.**

**Ans.** In the **lock and key model**, the **hormone acts as a key** that binds specifically to a **receptor (lock) on the target cell membrane or inside the cell.** Only cells with *the correct receptor respond to the hormone*, ensuring **specificity of action.** This mechanism explains how hormones trigger precise physiological responses in target tissues without affecting other cells.

> **MARKING SCHEME (3 Marks)**
> - Hormone specificity (1 mark), explanation of lock and key *model* (2 marks)

**Q.8 How does the pituitary gland control the thyroid gland?**

**Ans.** The anterior pituitary gland secretes thyroid-stimulating hormone (TSH), which binds to receptors on the thyroid gland, stimulating it to produce thyroxine (T4) and triiodothyronine (T3). These thyroid hormones regulate metabolism, growth, development, and energy expenditure, thus linking the pituitary's hormonal control to body-wide metabolic functions.

> **MARKING SCHEME (3 Marks)**
> - TSH and its role (2 marks), thyroxine and its role (1 mark)

**Q.9 What is endocrine-to-endocrine communication?**

**Ans.** Endocrine-to-endocrine communication occurs when **one endocrine gland regulates the activity of another gland** through hormone secretion. For example, **TSH from the pituitary gland stimulates the thyroid gland** to release thyroxine, which in turn affects metabolic processes. This mechanism ensures **coordinated regulation of multiple physiological systems.**

> **MARKING SCHEME (3 Marks)**
> - Definition (2 marks), example (1 mark)

> **SLO [B-12-G-25]**
> **Locate the endocrine glands in human body name the hormones they release and their functions; (pituitary, thyroid, parathyroid, pancreas, adrenal, gonads)**

## ENDOCRINE SYSTEM OF MAN

Endocrine system is the type of glandular system, consists of some 20 ductless glands lying in different parts of the body. Some of the major endocrine glands, their locations and hormonal secretions are shown in the given figure.

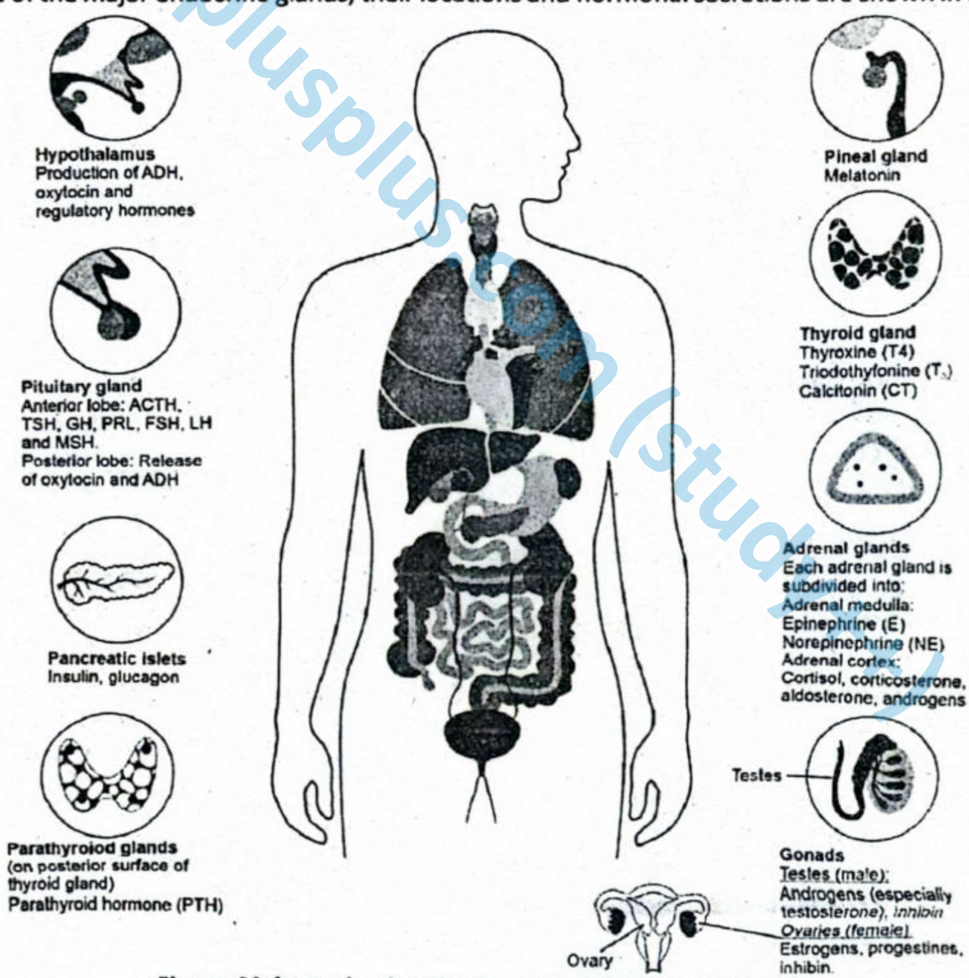



**Figure: Major endocrine glands and their locations in human**

## The Pituitary Gland: Master Regulator of the Endocrine System:

**Why called master gland:**

The **pituitary gland**, often referred to as the **"master gland"**, plays a central role in the regulation of the endocrine system. It is located at the base of the brain and is anatomically connected to the **hypothalamus**, which controls its secretory activity through releasing and inhibiting hormones.

The gland is divided into three functional lobes:

- Anterior lobe (adenohypophysis)
- Intermediate lobe (pars intermedia)
- Posterior lobe (neurohypophysis)

### ▸ Anterior Lobe (Adenohypophysis)

The anterior pituitary secretes **six major peptide hormones**, most of which regulate the activity of other endocrine glands. It is thus a key component in **hypothalamic-pituitary-target gland axes.**

1. **Growth Hormone (GH) / Somatotropic Hormone (STH)**
   - Secretion is stimulated by **growth hormone-releasing factor (GHRF)** and inhibited by **somatostatin**, both from the hypothalamus.
   - GH promotes **growth and development** by stimulating:
     - **Cell division and cellular growth**
     - **Amino acid uptake**
     - **Protein synthesis**

   **Disorders:**
   - **Deficiency** during childhood causes **pituitary dwarfism**, characterized by proportionate short stature with normal intelligence.
   - **Over-secretion** during childhood leads to **gigantism**, marked by excessive linear bone growth.
   - **Over-secretion in adults** results in **acromegaly**, where bones no longer elongate but thicken, causing enlargement of the hands, feet, skull, jaw, and facial features.

2. **Thyroid-Stimulating Hormone (TSH)**
   - Controlled by **thyrotropin-releasing factor (TRF)** from the hypothalamus.
   - TSH regulates the activity of the **thyroid gland**, promoting:
     - Increase **in size and number** of thyroid cells
     - **Synthesis and release** of thyroid hormones ($T_3$ and $T_4$)

   **Disorders:**
   - Over-secretion leads to hyperthyroidism.
   - Under-secretion results in hypothyroidism.

3. **Adrenocorticotropic Hormone (ACTH)**
   - Stimulated by **corticotropin-releasing factor (CRF)** from the hypothalamus, especially in response to **stress** (e.g., cold, pain, infection).
   - ACTH stimulates the **adrenal cortex** to release **corticosteroids**, including **cortisol** and **aldosterone.**

4. **Follicle-Stimulating Hormone (FSH)**
   - In **females**, FSH stimulates **follicular development** in the ovaries.
   - In **males**, it stimulates **spermatogenesis** in the seminiferous tubules.

5. **Luteinizing Hormone (LH) / Interstitial Cell-Stimulating Hormone (ICSH)**
   - In **females**, LH induces ovulation and promotes corpus luteum formation.
   - In **males**, ICSH stimulates testosterone secretion from interstitial (Leydig) cells in the testes.

6. **Prolactin (PRL) / Luteotropic Hormone (LTH)**
   - Promotes **milk production** (lactogenesis) in mammary glands following childbirth.
   - Its release is modulated by **prolactin-inhibiting hormone (dopamine)** from the hypothalamus.

### ▸ Intermediate Lobe (Pars Intermedia)

The **intermediate lobe** is poorly developed in humans and forms a thin layer between the anterior and posterior lobes. It secretes:

**Melanocyte-Stimulating Hormone (MSH):**

- Regulated by **MSH-inhibiting hormone** from the hypothalamus.
- Stimulates **melanin production** by melanocytes in the skin and hair.
- MSH secretion increases during **pregnancy**, contributing to pigmentation changes.

### ▶ Posterior Lobe (Neurohypophysis)

The posterior lobe is **not glandular**. It does not synthesize hormones but serves as a **storage and release site** for hormones produced by **neurosecretory cells of the hypothalamus**. These hormones are transported down axons and released into the bloodstream in response to neural signals.

**1. Antidiuretic Hormone (ADH) / Vasopressin**

- Released during dehydration, low blood volume, or hypotension.
- Promotes water reabsorption in the renal collecting ducts, thereby conserving water and maintaining blood pressure.

**Disorders:**

- Under-secretion causes diabetes insipidus, characterized by:
    - **Excessive dilute urine** (polyuria)
    - **Intense thirst** (polydipsia)
- **Over-secretion** may impair kidney function and contribute to **fluid retention** and **hyponatremia**.

**2. Oxytocin**

- Released in **pulses during childbirth**, stimulating **uterine contractions** to facilitate labor.
- **In nursing mothers**, oxytocin triggers **milk ejection** (let-down reflex) in response to infant suckling by stimulating contraction of myoepithelial cells in the mammary glands.

**Disorders:**

- Under-secretion may delay or inhibit labor and milk release.
- Excessive secretion during delivery may cause uterine rupture if not medically managed.

### ▶ Exocrine Glands

In contrast to endocrine glands, **exocrine glands** secrete substances through **ducts** onto body surfaces or into cavities. Examples include:

- Salivary glands (saliva)
- Sweat glands (perspiration)
- Mammary glands (milk)
- Lacrimal glands (tears)
- Digestive glands (enzymes)

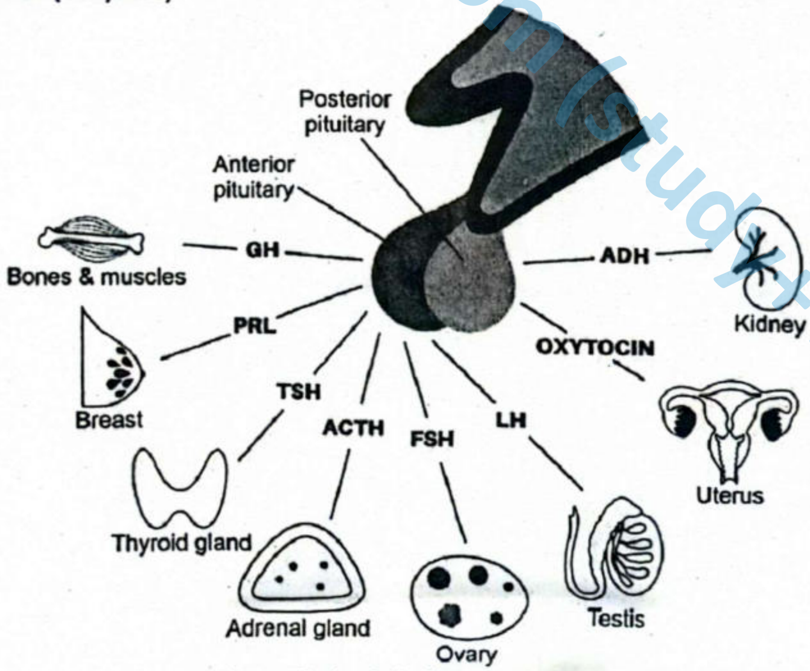



**Figure: Role of pituitary hormones**

## Thyroid Gland

### Thyroid Gland and Its Hormonal Functions

The thyroid gland is a bilobed endocrine organ situated on either side of the trachea, just inferior to the larynx. It consists of two lateral lobes connected by a narrow isthmus. The gland synthesizes and secretes three biologically active hormones: **triiodothyronine ($T_3$), thyroxine ($T_4$ or tetraiodothyronine)**, and **calcitonin.**

### Triiodothyronine ($T_3$) and Thyroxine ($T_4$)

$T_3$ and $T_4$ are iodine-containing hormones synthesized by follicular cells of the thyroid gland. Triiodothyronine ($T_3$) contains three iodine atoms, while thyroxine ($T_4$) contains four, hence their names. Their secretion is regulated by **thyroid-stimulating hormone (TSH)** from the anterior pituitary.

These hormones exert widespread physiological effects, including:

- Increasing the **basal metabolic rate** of cells throughout the body.
- Stimulating **glucose catabolism** and **cholesterol synthesis in the liver.**
- Supporting the **development and maturation of the nervous system**, particularly in the fetus and during infancy.
- Facilitating **muscle development and function.**
- Promoting **skeletal growth** and **maturation.**
- Enhancing **gastrointestinal motility** for effective digestion and absorption.

### Disorders Related to $T_3$ and $T_4$

### Hyperthyroidism:

**Definition:**

- Hyperthyroidism is a condition marked by excessive secretion of $T_3$ and $T_4$.

**Example:**

- One common form is **Graves' disease.** It is an autoimmune disorder characterized by increased metabolic rate, weight loss, irritability, and exophthalmos (protruding eyes).

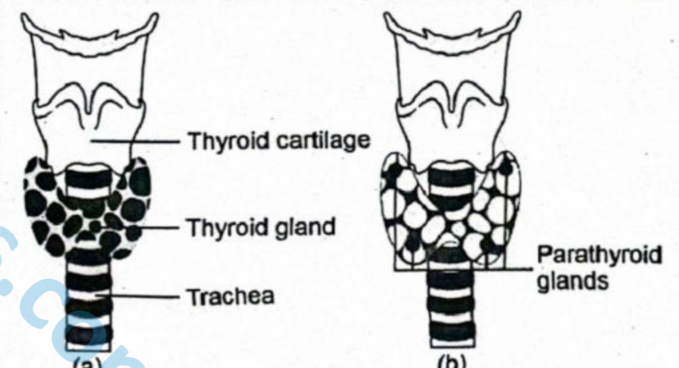


**Figure: Thyroid and parathyroid gland**

### Hypothyroidism:

**Definition:**

- It results from under-secretion of thyroxine.

**Example:**

- In adults, this condition is termed **myxedema.**

**Symptoms:**

- Reduced metabolic rate,
- sensitivity to cold,
- puffiness of the face (including periorbital edema),
- dry and thickened skin,
- thinning of hair (including loss from the scalp and eyebrows),
- constipation,
- Swelling of the tongue.

Myxedema may also cause **goiter**, an enlargement of the thyroid gland, often due to **iodine deficiency.**

### Cretinism:

- In infants and young children, congenital hypothyroidism causes a condition known as **cretinism,**

**Symptoms:**

- Characterized by severe mental retardation,
- Stunted physical growth,
- Disproportionate body size.
- Delayed bone maturation, late onset of puberty,
- Infertility are commonly observed in untreated individuals.

- **Calcitonin**

**Releases from:**

Calcitonin is a peptide hormone secreted by the **parafollicular (C) cells** of the thyroid gland in response to elevated blood calcium levels.

**Functions:**

Functions of calcitonin include:

- Enhancing **calcium deposition** in the bone matrix by stimulating osteoblastic activity.
- Inhibiting **intestinal absorption** of calcium.
- Reducing **renal reabsorption** of calcium, thereby promoting its excretion in the urine.
- Calcitonin plays a more prominent role during **childhood**, when skeletal growth is rapid, and bone remodeling is active.

**Deficiency:**

In cases of calcitonin deficiency, calcium may not be efficiently deposited into bones, leading to **hypercalcemia,** which can disrupt **neuromuscular function** and contribute to the formation of **kidney stones.**

## Parathyroid Glands and Calcium Regulation

- **Location:**

In humans, the parathyroid glands are **four small, oval-shaped endocrine glands** embedded in the posterior surface of the thyroid gland, typically arranged as two pairs—one pair on each lobe of the thyroid. These glands are usually light-colored and are crucial for calcium homeostasis.

- **Secretion:**

The principal hormone secreted by the parathyroid glands is parathormone (PTH), also known as parathyroid hormone. It is the single most critical regulator of blood calcium levels.

- Secretion of PTH is stimulated by low blood calcium concentrations and is inhibited when blood calcium levels are high.
- PTH acts antagonistically to calcitonin, the hormone secreted by the thyroid gland.

- **Functions of Parathormone**

Parathormone maintains blood calcium homeostasis through multiple mechanisms:

- **Bone:** PTH stimulates osteoclasts, increasing bone resorption and releasing calcium into the bloodstream.
- **Kidneys:** It enhances calcium reabsorption in the renal tubules and reduces phosphate reabsorption.
- **Intestines:** PTH indirectly promotes calcium absorption by stimulating the production of **calcitriol (active vitamin $D_3$)**, which enhances dietary calcium absorption from the gut.

- **Disorders Related to Parathormone**

**Over-secretion (Hyperparathyroidism)**

- Hyperparathyroidism is often caused by a **tumor of the parathyroid gland** (parathyroid adenoma).
- Excessive PTH causes **increased calcium release from bones**, leading to **bone demineralization, softening, and increased risk of spontaneous fractures.**
- Blood calcium levels become abnormally high (**hypercalcemia**), which may:
  - Depress the activity of the central nervous system, causing **fatigue and muscle weakness.**
  - Result in the **formation of kidney stones** due to the precipitation of excess calcium salts in the urinary tract.

**Under-secretion (Hypoparathyroidism)**

- Hypoparathyroidism results in **hypocalcemia**, a condition where blood calcium levels fall below normal.
- Low calcium levels **increase the excitability of neurons**, which may lead to **tetany**—a condition characterized by prolonged, involuntary muscle contractions.

## Pancreas: Dual Role in Digestion and Endocrine Regulation

The **pancreas** is a unique organ that performs both **exocrine** and **endocrine** functions.

- **Exocrine Function**

The majority of pancreatic tissue is exocrine in nature. This portion of the gland is composed of **acinar cells**, which produce and secrete **digestive enzymes** into the **duodenum** through the pancreatic duct. These enzymes are essential for the breakdown of carbohydrates, proteins, and lipids in the small intestine.

## Endocrine Function

Scattered throughout the exocrine tissue are clusters of specialized endocrine cells known as the **Islets of Langerhans**. These islets are responsible for the secretion of hormones that regulate **blood glucose levels.** There are two principal types of hormone-secreting cells within the islets:

- **Beta (β) cells:** More numerous, they secrete **insulin.**
- **Alpha (α) cells:** Fewer in number, they secrete **glucagon.**

Both cell types respond directly to fluctuations in blood glucose concentrations.

## Insulin

**Insulin** is secreted by β-cells in response to **elevated blood glucose levels**, such as those observed after a carbohydrate-rich meal.

### Functions of Insulin:

- **Lowers blood glucose** by facilitating glucose uptake into most body cells, especially **skeletal muscle and adipose tissue**.
- Stimulates ***glycogenesis*** (conversion of glucose to glycogen) in the liver and muscles.
- Enhances **cellular respiration** by promoting the utilization of glucose.
- Promotes the **conversion of excess glucose into lipids** for storage.
- Inhibits **gluconeogenesis**, the synthesis of glucose from non-carbohydrate sources.

### Disorders Associated with Insulin:

- **Under-secretion of insulin** leads to **diabetes mellitus,** a metabolic disorder characterized by **hyperglycemia** (elevated blood glucose), and the presence of **glucose in the urine** (glycosuria).
- **Excess insulin secretion** can result in **hypoglycemia**, where glucose levels fall below the normal range, potentially impairing **neuromuscular function**, leading to **confusion, weakness, and in severe cases, unconsciousness.**

## Glucagon

**Glucagon** is secreted by α-cells when **blood glucose levels fall**, such as during fasting or between meals. Its secretion is also stimulated by the **sympathetic nervous system**. Conversely, high glucose levels, insulin, and somatostatin inhibit glucagon release.

### Functions of Glucagon:

- **Increases blood** glucose by stimulating:
  - **Glycogenolysis** (breakdown of glycogen to glucose) in the liver.
  - **Gluconeogenesis** (formation of glucose from non-carbohydrate sources).
- Acts **antagonistically to insulin**, counteracting its glucose-lowering effects and maintaining glucose homeostasis.

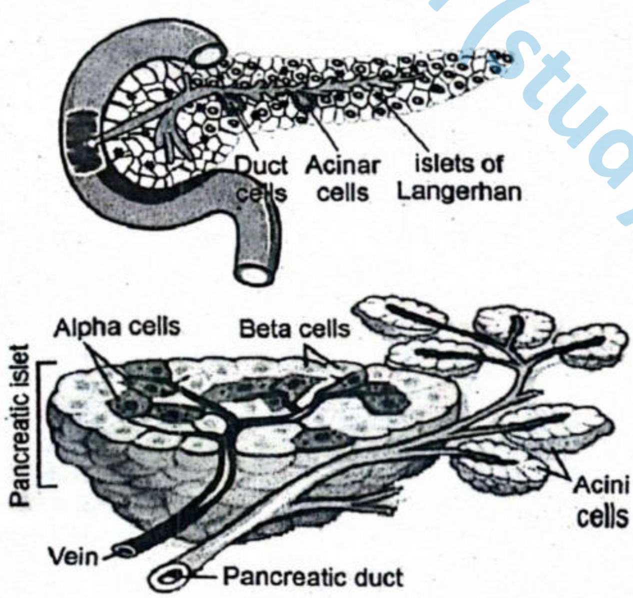



**Figure: Pancreas and Islets of Langerhans**

## Adrenal Glands and Their Hormonal Functions

- **Location:**

The **adrenal glands** are a pair of endocrine glands located **on top of each kidney,** hence also known as **suprarenal glands.** Each adrenal gland consists of two distinct regions:

- The **adrenal medulla** (inner region)
- The **adrenal cortex** (outer region)

These regions differ in structure, function, and embryological origin, and they secrete different classes of hormones.

- **Adrenal Medulla**

The adrenal medulla is composed of **chromaffin cells,** which are modified sympathetic neurons. It functions as part of the **sympathoadrenal system,** secreting **catecholamines** in response to acute stress, particularly under stimulation by the **sympathetic nervous system.**

**Hormones of the Adrenal Medulla:**

- Epinephrine (adrenaline)
- Norepinephrine (noradrenaline)

These hormones are secreted during **emergency or "fight-or-flight" situations.** They prepare the body for immediate physical action by producing a **synergistic effect**—both hormones target similar systems but differ in their intensity and specific actions.

- **Epinephrine** is the more potent stimulator of:
  - Metabolic rate (increased glucose availability)
  - Bronchodilation (widening of airways)
  - Increased blood flow to **skeletal muscles** and the **heart**
- **Norepinephrine** exerts a stronger effect on:
  - **Peripheral vasoconstriction,** leading to increased systemic vascular resistance

The combined result is a **rise in heart rate, respiratory rate, blood pressure,** and energy availability to meet the demands of stress.

**Disorders Related to Medullary Hormones:**

- **Over-secretion** may lead to persistent hypertension, tachycardia, and aggressive behavior, even in the absence of stress stimuli.
- **Under-secretion** impairs the body's ability to cope with acute stress, resulting in reduced emergency responsiveness.

- **Adrenal Cortex**

The adrenal cortex produces **steroid hormones,** collectively known as **corticosteroids,** under the regulation of **adrenocorticotropic hormone (ACTH)** from the anterior pituitary.

**Major Hormonal Classes:**

**(i) Glucocorticoids** (e.g., **cortisol, cortisone**):
  - Regulate **carbohydrate, protein, and lipid metabolism.**
  - Maintain **blood glucose levels** during fasting or stress.
  - Suppress **inflammatory** and **immune responses.**

**(ii) Mineralocorticoids (e.g., aldosterone):**
  - Regulate the balance of **sodium, potassium, and water.**
  - Promote sodium reabsorption and potassium excretion in the kidneys.
  - Play a critical role in **blood volume and pressure regulation.**

**(iii) Adrenal androgens** (e.g., small amounts of **testosterone**):
  - Contribute to the development of **secondary sexual characteristics,** especially in females.
  - In males, their role is minimal due to the dominant production of testosterone by the testes.

**Disorders Related to Cortical Hormones:**

- ***Under-secretion*** of adrenal cortical hormones causes **Addison's disease,** characterized by:
  - General **metabolic disturbances.**
  - **Muscle weakness.**
  - **Electrolyte imbalance,** especially sodium loss.
  - Extreme sensitivity to stress, such as exposure to cold, which may result in **collapse or death.**

- **Over-secretion**, particularly of glucocorticoids, leads to **Cushing's syndrome**, marked by:
    - Excessive **protein catabolism.**
    - **Muscle wasting** and **bone demineralization.**
    - Redistribution of body fat (e.g., "moon face" and "buffalo hump").
    - Suppression of the immune system.

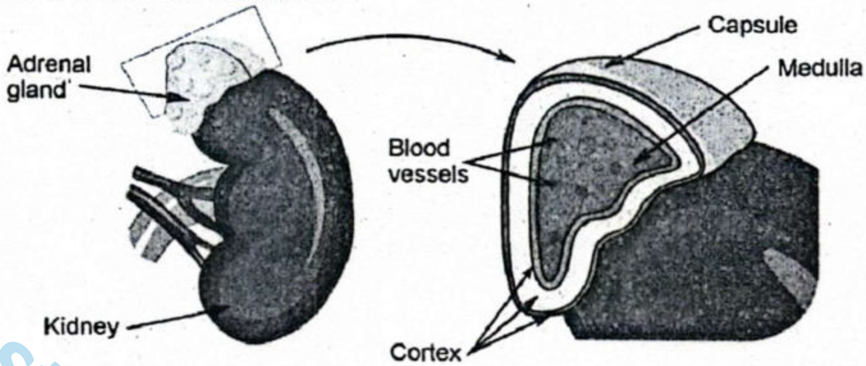



**Figure: Adrenal gland**

## Gonads (Testes and Ovaries)

### Testes:

The testes are the male reproductive organs responsible for the production of sperm and the secretion of the male sex hormone, testosterone. Testosterone is synthesized by the interstitial (Leydig) cells located between the seminiferous tubules, under the influence of interstitial cell-stimulating hormone (ICSH), a gonadotropin secreted by the anterior pituitary.

- During puberty, testosterone promotes the maturation of male reproductive organs and the emergence of secondary sexual characteristics such as increased facial and body hair, deepening of the voice, and greater muscle mass.
- It plays a central role in initiating and maintaining the male libido (sex drive).
- Testosterone is essential for the process of spermatogenesis and the maintenance of the male reproductive system in its fully functional adult state.
- **Under-secretion of testosterone** may lead to the development of feminized features, reduced libido, impaired sperm production, and male infertility.

### Ovaries (Female Gonads)

The ovaries are the female reproductive organs that produce ova and secrete the sex hormones estrogen and progesterone.

- **Estrogen** is primarily responsible for the development of female secondary sexual characteristics such as breast development, widening of the pelvis, and the distribution of body fat. It also regulates the menstrual cycle by promoting the proliferation of the uterine lining (endometrium).
- **Progesterone** plays a vital role in preparing the uterus for implantation of a fertilized egg and in maintaining pregnancy.

### Role of Progesterone

- Progesterone is secreted by the corpus luteum in response to luteinizing hormone (LH) during the second half of the menstrual cycle.
- In pregnancy, the placenta becomes the major source of progesterone.
- This hormone thickens and vascularizes the uterine lining to support implantation and sustains the pregnancy.
- It *also* suppresses ovulation by inhibiting the secretion of follicle-stimulating hormone (FSH), preventing the maturation of additional follicles during the pregnancy cycle.
- **Under-secretion of progesterone during the menstrual cycle** may result in insufficient endometrial development, reducing the chances of implantation and potentially leading to *early menstruation.*
- **Under-secretion during pregnancy** may cause failure to maintain the uterine lining, increasing the risk of miscarriage.

## Pineal Gland

- The pineal gland is a small structure located in the brain.

- It secretes **melatonin**, which regulates circadian rhythms and sleep-wake cycles. Its production is influenced by light exposure.

## Thymus Gland

- The thymus is located behind the sternum and is more prominent in children.
- It secretes **thymosins,** which play a role in the development and differentiation of T lymphocytes, essential for adaptive immunity.

**SCIENCE TITBITS**

- Under certain conditions, such as severe blood loss, exceptionally large amounts of ADH are released, causing a rise in blood pressure. The alternative name for this hormone, vasopressin, reflects this particular effect.
- $T_4$ (also known as thyroxin) is the major hormone, about 90%, secreted by the thyroid; $T_3$ is only 10% $T_3$ is four times more potent than $T_4$; however, action duration of $T_4$ is four times more than $T_3$.
- Graves' disease is believed to be an autoimmune disease. The serum of patients contains abnormal antibodies that mimic TSH and continuously stimulate thyroxin release. The symptoms include high metabolic rate, rapid and irregular heartbeat, increased breathing rate, increased body temperature, sweating and weight loss despite adequate food intake. Mostly exophthalmia (protrusion of the eyeballs) results from Graves's disease and is a classic symptom of hyperthyroidism.
- Hormones released from kidney
- Renin monitors blood pressure and takes corrective action if it drops.
- Erythropoietin acts on the bone marrow to increase the production of red blood cells. Stimuli such as bleeding or moving to high altitudes (where oxygen is scarcer) trigger the release of EPO.
- Calcitriol acts on the cells of the intestine to promote the absorption of calcium from the diet.
- Pancreatic acinar cells are functional units of the exocrine pancreas. They synthesize, store, and secrete inactive digestive enzymes into the lumen of the acinus.

**Table: Summary of Endocrine Glands in Humans**

| Endocrine Gland | Location | Hormones Secreted | Major Functions |
|---|---|---|---|
| **Pituitary Gland (Anterior lobe)** | Base of brain, below hypothalamus | • Growth Hormone (GH) – Thyroid<br>• Stimulating Hormone (TSH)<br>• Adrenocorticotropic Hormone (ACTH)<br>• Follicle-Stimulating Hormone (FSH)<br>• Luteinizing Hormone (LH/ICSH)<br>• Prolactin (PRL) | • Stimulates body growth and protein synthesis<br>• Regulates thyroid gland activity<br>• Stimulates adrenal cortex<br>• Controls gamete production and gonadal hormone secretion<br>• Promotes milk production |
| **Pituitary Gland (Intermediate lobe)** | Between anterior and posterior lobes | • Melanocyte-Stimulating Hormone (MSH) | • Stimulates melanin production in skin and hair |
| **Pituitary Gland (Posterior lobe)** | Base of brain, stores hormones from hypothalamus | • Antidiuretic Hormone (ADH) – Oxytocin | • Promotes water reabsorption in kidneys<br>• Triggers uterine contractions and milk ejection |
| **Thyroid Gland** | In the neck, around trachea below larynx | • Triiodothyronine ($T_3$)<br>• Thyroxine ($T_4$)<br>• Calcitonin | • Increases metabolic rate<br>• Supports nervous system development<br>• Regulates blood calcium by promoting deposition in bones |

| | | | |
|---|---|---|---|
| **Parathyroid Glands (4)** | Posterior surface of thyroid gland | • Parathyroid Hormone (PTH) | • Increases blood calcium levels<br>• Stimulates bone resorption, calcium absorption from intestines, and reabsorption by kidneys |
| **Adrenal Glands (Medulla)** | Inner part of adrenal gland, on top of each kidney | • Epinephrine (adrenaline)<br>• Norepinephrine (noradrenaline) | • Initiates "fight-or-flight" response<br>• Increases heart rate, blood pressure, respiration rate, and glucose release |
| **Adrenal Glands (Cortex)** | Outer layer of adrenal gland, on top of each kidney | • Glucocorticoids (e.g., cortisol)<br>• Mineralocorticoids (e.g., aldosterone)<br>• Androgens (e.g., testosterone in small amounts) | • Regulate metabolism and immune response<br>• Control electrolyte and water balance<br>• Contribute to secondary sexual characteristics (minor role) |
| **Pancreas (Islets of Langerhans)** | Behind the stomach | • Insulin (from β-cells)<br>• Glucagon (from α-cells) | • Insulin lowers blood glucose by promoting uptake into cells<br>• Glucagon raises blood glucose by *stimulating glycogen* breakdown in the liver |
| **Ovaries (in females)** | Pelvic cavity | • Estrogen<br>• Progesterone | • Regulate menstrual cycle<br>• Promote development of female secondary sexual characteristics<br>• Maintain pregnancy |
| **Testes (in males)** | Scrotum | • Testosterone | • Stimulates development of male secondary sexual characteristics<br>• Promotes sperm production |
| **Pineal Gland** | Deep in the brain, near the center, above the midbrain | • Melatonin | • Regulates circadian rhythms (sleep-wake cycle)<br>• Influences seasonal biological cycles |
| **Thymus Gland** | Upper chest, behind the sternum, between lungs | • Thymosins | • Stimulate maturation and differentiation of T lymphocytes<br>• Support immune function during early life |

SLO [B-12-G-26]

**Relate problems associated with the imbalance of these hormones.**

**Table: Disorders Associated with Human Endocrine Glands**

| Endocrine Gland | Disorder | Description |
|---|---|---|
| **Anterior Pituitary** | Dwarfism | Under-secretion of growth hormone (GH) during childhood results in short stature with normal body proportions and mental development. |

| | | |
|---|---|---|
| | Gigantism | Over-secretion of GH during childhood causes abnormal increase in height before bone growth plates close. |
| | Acromegaly | Over-secretion of GH in adulthood leads to thickening of bones, especially in the face, hands, and feet. |
| | Hyperthyroidism | Excess TSH stimulates thyroid gland to overproduce thyroxine, causing symptoms like weight loss and rapid heart rate. |
| | Hypothyroidism | Inadequate TSH secretion reduces thyroxine levels, leading to fatigue, weight gain, and cold intolerance. |
| **Posterior Pituitary** | Diabetes insipidus | Under-secretion of antidiuretic hormone (ADH) leads to excessive dilute urine and persistent thirst. |
| | Kidney dysfunction | Over-secretion of ADH may cause water retention and interfere with kidney function. |
| | Impaired labour | Deficient oxytocin secretion disrupts normal uterine contractions during childbirth. |
| | Uterine rupture | Excess oxytocin may cause dangerously strong contractions and rupture of the uterine wall. |
| **Thyroid Gland** | Hyperthyroidism | Overproduction of thyroxine causes increased metabolic rate, weight loss, and nervousness. |
| | Hypothyroidism | Deficient thyroxine secretion leads to slow metabolism, weight gain, and fatigue. |
| | Goitre | Abnormal enlargement of the thyroid gland, often due to iodine deficiency. |
| **Parathyroid Glands** | Hypercalcemia | Over-secretion of parathyroid hormone (PTH) increases blood calcium, weakening bones and causing kidney stones. |
| | Tetany | Under-secretion of PTH results in low calcium levels, leading to muscle spasms and cramps. |
| **Adrenal Cortex** | Addison's disease | Under-secretion of cortisol and aldosterone causes fatigue, hypotension, and electrolyte imbalance. |
| | Cushing's syndrome | Excess cortisol leads to central obesity, high blood sugar, and muscle weakness. |
| | Hyperaldosteronism | Overproduction of aldosterone results in hypertension and low potassium levels. |
| **Adrenal Medulla** | Pheochromocytoma | Tumor causing overproduction of adrenaline and noradrenaline, leading to episodic hypertension. |
| | Persistent hypertension | Sustained high blood pressure due to excessive catecholamine secretion. |
| **Pancreas (Islets of Langerhans)** | Diabetes mellitus | Insufficient insulin or insulin resistance causes chronic high blood glucose levels. |
| | Hypoglycemia | Excessive insulin lowers blood sugar to dangerous levels, causing confusion, weakness, or coma. |
| **Ovaries** | Irregular menstrual cycles | Hormonal imbalance in estrogen and progesterone disrupts menstrual regulation. |
| | Infertility | Failure to ovulate due to insufficient gonadotropic hormone secretion. |
| | Polycystic ovary syndrome (PCOS) | Hormonal disorder with ovarian cysts, irregular menstruation, and elevated androgens. |
| **Testes** | Delayed puberty | Low testosterone secretion delays onset of secondary sexual characteristics. |

| | | |
|---|---|---|
| | Infertility | Reduced sperm production due to hormonal imbalance affects fertility. |
| | Underdeveloped male features | Deficiency of testosterone results in poor development of male physical traits. |
| **Pineal Gland** | Sleep disorders | Imbalance in melatonin secretion disrupts circadian rhythms, leading to insomnia or excessive sleepiness. |
| | Seasonal affective disorder (SAD) | Reduced melatonin production during winter months may contribute to depressive symptoms. |
| **Thymus Gland** | Immunodeficiency | Underactivity of thymus impairs T cell maturation, weakening immune defense especially in early life. |
| | Autoimmune conditions | Thymic dysfunction may cause the immune system to attack the body's own tissues. |

## SLOs Based (MCQs)

## ENDOCRINE SYSTEM OF MAN



**1. Which hormone is responsible for stimulating body growth and cell division?**
A. TSH B. GH
C. ACTH D. ADH

**2. Which hormone increases melanin production in the skin and hair?**
A. MSH B. ACTH
C. FSH D. TSH

**3. Which gland is referred to as the "master gland"?**
A. Thyroid B. Hypothalamus
C. Adrenal gland D. Anterior pituitary

**4. Gigantism is caused by:**
A. Over-secretion of thyroxine
B. Over-secretion of GH in childhood
C. Under-secretion of GH in childhood
D. Over-secretion of ACTH

**5. What is the primary function of ADH?**
A. Stimulates ovulation
B. Promotes water reabsorption in kidneys
C. Increases metabolic rate
D. Triggers labor contractions

**6. Hypocalcemia is a result of:**
A. Over-secretion of PTH
B. Under-secretion of calcitonin
C. Under-secretion of PTH
D. Excess insulin

**7. Diabetes insipidus is caused by deficiency of:**
A. ADH B. Insulin
C. Cortisol D. TSH

**8. Which gland produces hormones involved in the fight-or-flight response?**
A. Pancreas B. Adrenal medulla
C. Thyroid D. Pineal gland

**9. Over-secretion of parathormone leads to:**
A. Tetany B. Acromegaly
C. Hypercalcemia D. Hypoglycemia

**10. The hormone oxytocin is released in response to:**
A. Stress B. Dehydration
C. Suckling D. Low calcium levels

**11. Which of the following glands secretes melatonin?**
A. Thymus B. Adrenal
C. Pineal D. Pancreas

**12. Which hormone increases the metabolic rate of the body?**
A. ADH B. $T_4$
C. LH D. PRL

**13. Under-secretion of insulin causes:**
A. Tetany B. Diabetes mellitus
C. Diabetes insipidus D. Addison's disease

**14. ACTH stimulates:**
A. Adrenal cortex B. Adrenal medulla
C. Pineal gland D. Pancreas

**15. Which of the following is a gonadotropic hormone?**
A. TSH B. FSH
C. ADH D. Calcitonin

**16. Which hormone is associated with the development of secondary sexual characteristics in males?**
A. LH B. FSH
C. Testosterone D. ACTH

**17. Cretinism results from under-secretion of:**
A. GH B. $T_4$
C. Insulin D. ADH

**18. What is the function of prolactin?**
A. Stimulate ovulation
B. Promote milk production
C. Control calcium levels
D. Regulate adrenal cortex

19. **The thymus gland is involved in:**
A. Releasing digestive enzymes
B. Regulating calcium
C. Maturation of T-lymphocytes
D. Metabolism

20. **Acromegaly is due to:**
A. Under-secretion of GH in adults
B. Over-secretion of GH in adults
C. Under-secretion of insulin
D. Over-secretion of thyroxine

21. **Addison's disease results from:**
A. Over-secretion of adrenal cortex hormones
B. Deficiency of insulin
C. Under-secretion of adrenal cortex hormones
D. Over-secretion of PTH

22. **The hormone calcitonin is secreted by:**
A. Parathyroid gland B. Thyroid gland
C. Pituitary gland D. Adrenal cortex

23. **Which endocrine disorder involves spontaneous fractures and kidney stones?**
A. Diabetes mellitus B. Acromegaly
C. Hyperparathyroidism D. Addison's disease

24. **Which hormone triggers labor contractions?**
A. Oxytocin B. FSH
C. LH D. PRL

25. **Which hormone is regulated by melatonin production in a circadian rhythm?**
A. ADH B. Cortisol
C. Melatonin D. $T_3$

26. **A hormone that increases blood calcium level is:**
A. Calcitonin B. PTH
C. TSH D. Cortisol

27. **Graves' disease is caused by:**
A. Over-secretion of insulin
B. Over-secretion of thyroxine
C. Under-secretion of cortisol
D. Over-secretion of testosterone

28. **Which hormone regulates water balance in the body?**
A. Oxytocin B. ADH
C. LH D. Cortisol

29. **Tetany results from:**
A. Low levels of thyroxine B. Low calcium levels
C. Excess growth hormone D. High blood sugar

30. **The thymus gland secretes:**
A. TSH B. Thymosin
C. Oxytocin D. Cortisol

## Answer Key

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. B | 2. A | 3. D | 4. B | 5. B | 6. C | 7. A | 8. B | 9. C | 10. C |
| 11. C | 12. B | 13. B | 14. A | 15. B | 16. C | 17. B | 18. B | 19. C | 20. B |
| 21. C | 22. B | 23. C | 24. A | 25. C | 26. B | 27. B | 28. B | 29. B | 30. B |

## Short Questions

**Q.1 Why is the anterior pituitary called the "master gland"?**

**Ans.** The anterior pituitary is called the **master gland** because it secretes multiple hormones that regulate the activity of other endocrine glands, thus controlling overall endocrine function. Key hormones include **Growth Hormone (GH)**, which stimulates body growth; **Thyroid-Stimulating Hormone (TSH)**, which controls thyroid hormone production; **Adrenocorticotropic Hormone (ACTH)**, which regulates adrenal cortex activity; **Follicle-Stimulating Hormone (FSH)** and **Luteinizing Hormone (LH)**, which control reproductive organs; and **Prolactin (PRL)**, which stimulates milk production. Through these hormones, the anterior pituitary indirectly influences metabolism, growth, stress response, and reproductive processes, establishing its central role in endocrine coordination and homeostasis.

> **MARKING SCHEME (3 Marks)**
> - Definition (1 mark), names of hormones (2 marks)

**Q.2 What is the function of growth hormone and what disorders are linked to its imbalance?**

**Ans.** Growth hormone (GH), secreted by the anterior pituitary, **stimulates cell growth, division, and protein synthesis**, promoting overall body growth, muscle development, and bone elongation. It also influences **metabolic activities**, including fat breakdown and glucose regulation.

> **MARKING SCHEME (3 Marks)**
> - Function (1 mark), disorders (2 marks)

**Deficiency of GH** in childhood results in **dwarfism**, characterized by reduced stature and delayed development. Conversely, **over-secretion in children** causes **gigantism**, with excessive height and body size, while in adults it leads to **acromegaly**, marked by enlarged hands, feet, and facial features. Proper GH levels are critical for normal growth, metabolism, and tissue repair.

**Q.3 Describe the role of ADH in the human body.**

**Ans.** Antidiuretic hormone (ADH), produced by the posterior pituitary, **promotes water reabsorption in the kidney's collecting ducts,** reducing urine volume and preventing dehydration. It helps maintain **blood osmolarity and fluid balance.** ADH secretion is triggered by high blood osmolarity or low blood volume. **Deficiency of ADH** causes **diabetes insipidus**, characterized by excessive urination (polyuria), extreme thirst (polydipsia), and risk of dehydration. Overproduction of ADH can lead to **water retention, hyponatremia, and edema.** Through precise regulation of water balance, ADH ensures homeostasis, stabilizing blood pressure and osmotic conditions critical for cell function.

**MARKING SCHEME (3 Marks)**
- Function (1.5 marks), imbalance disorder (1.5 marks)

18
18

**Q.4 How does parathormone regulate calcium levels in the blood?**

**Ans.** Parathormone (PTH), secreted by the parathyroid glands, **raises blood calcium levels** through multiple mechanisms. It stimulates **osteoclast activity**, breaking down bone to release calcium. It enhances **intestinal calcium absorption** by activating vitamin D, and **reduces renal calcium excretion**, conserving calcium in the blood. PTH also indirectly affects phosphate balance by increasing phosphate excretion. **Deficiency of PTH** can cause **hypocalcemia**, leading to muscle cramps and tetany, while excess PTH leads to **hypercalcemia**, causing weakened bones, kidney stones, and abnormal heart rhythms. By finely tuning calcium levels, PTH is vital for **neuromuscular function, bone integrity, and blood clotting.**

**MARKING SCHEME (3 Marks)**
- Function (1.5 marks), imbalance disorder (1.5 marks)

**Q.5 What happens in the body during hyperparathyroidism?**

**Ans.** Hyperparathyroidism occurs when excess parathormone (PTH) is secreted, causing **elevated blood calcium levels.** This leads to **bone softening and increased risk of fractures** due to excessive calcium resorption. **Muscle weakness and fatigue** occur as calcium imbalances affect neuromuscular function. Additionally, calcium deposition in the kidneys can cause **kidney stones,** and hypercalcemia may result in **abdominal pain, nausea, and cognitive disturbances.** Chronic hyperparathyroidism can lead to osteoporosis, hypertension, and cardiovascular complications. Early diagnosis and treatment are essential to prevent severe skeletal and renal complications.

**MARKING SCHEME (3 Marks)**
- 1 mark each for three main effects

**Q.6 Describe the symptoms of hypothyroidism in children.**

**Ans.** Hypothyroidism in children is marked by **cretinism**, a condition arising from insufficient thyroid hormone secretion. Key symptoms include **stunted growth**, resulting in short stature and delayed skeletal development; **mental retardation**, causing cognitive impairment, learning difficulties, and delayed language acquisition; and **delayed physical development**, such as late eruption of teeth, slow reflexes, and underdeveloped muscles. Additional signs may include dry skin, puffy face, a protruding tongue, and lethargy. Early diagnosis and thyroid hormone replacement therapy are crucial to prevent irreversible developmental delays. Monitoring hormone levels during infancy ensures normal physical and mental growth, highlighting the thyroid's essential role in childhood development.

**MARKING SCHEME (3 Marks)**
- 1 mark each for three main symptoms

**Q.7 What are the functions of the adrenal medulla hormones?**

**Ans.** The adrenal medulla secretes **epinephrine (adrenaline)** and **norepinephrine (noradrenaline),** catecholamine hormones that regulate the body's response to stress. They initiate the **fight-or-flight response** by **increasing heart rate**, ensuring adequate blood supply to muscles. They **raise blood pressure** through vasoconstriction, supporting circulatory demands during emergencies. They also **stimulate glucose release from the liver,** providing immediate energy for heightened physical activity. Additionally, these hormones enhance respiratory rate, pupil dilation, and alertness, preparing the body to respond rapidly to threats. Their coordinated action allows quick adaptation to stressors, *maintaining homeostasis during acute challenges.*

**MARKING SCHEME (3 Marks)**
- 1 mark each for three main functions

**Q.8 How do insulin and glucagon work antagonistically?**

**Ans.** Insulin and glucagon, secreted by the pancreas, maintain **blood glucose homeostasis** through opposing actions. **Insulin,** released by β-cells, **lowers blood glucose** by stimulating glucose uptake in muscle and adipose tissues and promoting glycogen synthesis in the liver. **Glucagon**, secreted by α-cells, **raises blood glucose** by stimulating glycogen breakdown (glycogenolysis) and glucose production (gluconeogenesis) in the liver. Their antagonistic interplay ensures stable glucose levels between meals and during fasting. A proper balance prevents hyperglycemia or hypoglycemia, supporting energy supply for cells, particularly the brain. Dysregulation can lead to diabetes mellitus or other metabolic disorders.

> **MARKING SCHEME (3 Marks)**
> - Role of insulin (1.5 marks), role of glucagon (1.5 marks)

**Q.9 Describe the role of oxytocin in childbirth and lactation.**

**Ans.** Oxytocin, secreted by the posterior pituitary, plays a critical role in **childbirth and lactation**. During labor, it **stimulates uterine contractions**, facilitating the progressive dilation of the cervix and delivery of the fetus. Its pulsatile release is enhanced by cervical stretching, creating a positive feedback loop. In lactation, oxytocin **triggers milk ejection** from mammary alveoli in response to infant suckling, enabling efficient breastfeeding. It also promotes maternal bonding and may reduce postpartum bleeding by contracting uterine muscles. Oxytocin's dual role ensures successful parturition and nourishment of the newborn, illustrating the hormone's essential functions in reproduction.

> **MARKING SCHEME (3 Marks)**
> - Role in childbirth (1.5 marks), role in lactation (1.5 marks)

**Q.10 What is the function of melatonin and where is it secreted from?**

**Ans.** Melatonin is a hormone **secreted by the pineal gland** in response to darkness, regulating the **sleep-wake cycle** or circadian rhythm. It **promotes restful sleep** by signaling to the body that it is nighttime, thereby influencing body temperature, hormone secretion, and alertness. Melatonin also has **antioxidant properties**, supports immune function, and may play a role in reproductive timing in seasonal breeders. Its secretion decreases with light exposure and increases in darkness, coordinating daily physiological rhythms. Proper melatonin levels are essential for healthy sleep patterns, mental alertness, and overall circadian balance.

> **MARKING SCHEME (3 Marks)**
> - Function (2 marks), location (1 mark)

> **SLO [B-12-G-27]**
> **Explain the neurosecretory role of hypothalamus.**

## NEUROSECRETORY ROLE OF THE HYPOTHALAMUS

The **hypothalamus** is a small but critical region of the brain, located below the **thalamus** and forming the floor of the **third ventricle**. It serves as a **link between the nervous and endocrine systems**, integrating neural signals and hormonal responses to regulate **homeostasis**, growth, reproduction, and stress responses.

### Neurosecretory Cells of the Hypothalamus

Specialized **neurons**, called **neurosecretory cells**, in the hypothalamus produce **hormones** that influence the activity of the **pituitary gland**. Unlike typical neurons that transmit electrical impulses, these cells **synthesize and release chemical messengers** directly into the bloodstream, functioning as both **neurons and endocrine cells**.

- **Hypothalamic Hormones and Their Functions**

The hypothalamus releases two main types of hormones: **releasing/inhibiting hormones** and **directly secreted neurohormones**.

1. **Releasing and Inhibiting Hormones:**
   These hormones are secreted into the **hypophyseal portal system** to control the **anterior pituitary gland**. Examples include:
   - **Thyrotropin-releasing hormone (TRH):** Stimulates secretion of **thyroid-stimulating hormone (TSH)**.
   - **Corticotropin-releasing hormone (CRH):** Stimulates **adrenocorticotropic hormone (ACTH)** release.
   - **Growth hormone-releasing hormone (GHRH)** and **Growth hormone-inhibiting hormone (GHIH or somatostatin):** Regulate **growth hormone** secretion.
   - **Gonadotropin-releasing hormone (GnRH):** Stimulates **FSH and LH** release, controlling reproductive functions.
2. **Directly Secreted Neurohormones:**
   *Some* hypothalamic hormones are stored in the **posterior pituitary** and released into the **bloodstream** as needed. Examples include:
   - **Antidiuretic hormone (ADH):** Regulates water balance by acting on the kidneys.
   - **Oxytocin:** Stimulates **uterine contractions** during childbirth and **milk ejection** during lactation.

## Significance of the Neurosecretory Function

Through its **neurosecretory role**, the hypothalamus acts as the **master regulator of the endocrine system**. It ensures that hormonal secretions are **coordinated with neural signals**, allowing the body to respond to internal and external stimuli effectively. This integration is essential for maintaining **homeostasis**, regulating **growth and metabolism**, controlling **reproductive functions**, and managing **stress responses**.

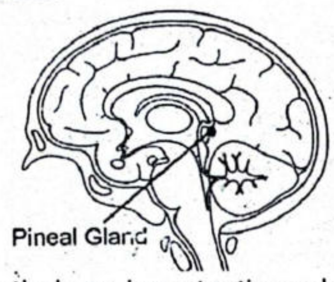
The hypothalamus, through its **neurosecretory cells**, bridges the nervous and endocrine systems. By releasing both **regulatory and direct hormones**, it controls the **pituitary gland** and indirectly governs many other endocrine glands, making it a central coordinator of the body's **chemical communication network**.

**Table 6.2 Hypothalamic hormones and their effect on pituitary gland**

| Hormone from the hypothalamus | Anterior pituitary response |
|---|---|
| Growth hormone releasing factors (GHRF) | Secretion of growth hormone (GH) |
| Somatostatin | Inhibition of GH |
| Thyrotrophin releasing factor (TRF) | Secretion of thyroid stimulating hormone (TSH) |
| Adrenocorticotrophin releasing factor (CRF) | Secretion of adrenocorticotrophic hormone (ACTH) |
| Prolactin inhibiting factor (PIF) | Inhibits secretion of prolactin |
| Gonadotrophin releasing hormone (GnRH) | Secretion of FSH and LH |

**SCIENCE TITBITS**

The **pineal gland** is a small, pea-shaped endocrine organ located deep within the brain, attached to the roof of the third ventricle near the **hypothalamus**. Its principal hormone is **melatonin**, which plays a key role in regulating the **circadian rhythm**—the body's internal clock that controls sleep-wake cycles and other daily physiological processes.




The **thymus gland**, situated behind the sternum and between the lungs, is most active and reaches its maximum size during **childhood**. It secretes a group of hormones collectively referred to as **thymosins**, which are essential for the development of the immune system. **Immature lymphocytes**, produced in the bone marrow, migrate to the thymus, where thymosin facilitates their differentiation into **T lymphocytes (T cells)**—a vital component of adaptive immunity.

**SCIENCE TITBITS**

**Other Endocrine Tissues/Cells**

Apart from classical endocrine glands, various other organs and tissues also produce hormones, although their primary function may not be endocrine in nature. These include the following:

- **Gastrointestinal Hormones**:
  - *Gastrin* is secreted by the stomach wall. It enhances the secretion of pepsinogen and hydrochloric acid, aiding in digestion.
  - *Secretin* and *cholecystokinin (CCK)* are produced by the epithelial cells of the duodenum. These hormones regulate pancreatic enzyme secretion and stimulate bile release from the liver and gallbladder.
- **Placenta**:
  The placenta, a temporary organ formed during pregnancy, acts as an endocrine gland by secreting hormones such as *progesterone*, which is crucial for the maintenance of pregnancy.
- **Prostaglandins**:
  Prostaglandins are a group of locally acting lipid-based hormones synthesized in various tissues. They are involved in diverse physiological functions including inflammation, pain regulation, smooth muscle contraction, and protection during infections.
- **Endorphins**:
  These peptide hormones are produced by neurons in the brain and spinal cord. Endorphins bind to opioid receptors and inhibit pain perception, thereby acting as natural analgesics.

**SLO [B-12-G-28]**
**Outline the concept of Feedback mechanism of hormones. Describe positive feedback with reference to oxytocin and negative feedback with reference to insulin and glucagon.**

## FEEDBACK MECHANISM OF HORMONES

**Feedback mechanisms** are fundamental control processes in biological systems, where the **output or product of a process regulates the process itself**. These mechanisms operate by detecting deviations from a **set point** and initiating responses to either **correct** or **reinforce** the change. In the endocrine system, feedback mechanisms are essential for maintaining **homeostasis** and regulating hormone levels. There are two primary types: **negative feedback**, which opposes the change, and **positive feedback**, which amplifies it.

### Positive Feedback Mechanism

**Positive feedback** occurs when the **end product of a process enhances or accelerates its own production.** Unlike negative feedback, positive feedback responses are typically **not homeostatic** and are less common, usually operating in processes that require a **definitive, self-amplifying outcome.**

- **Example: Childbirth and Oxytocin**

During **labor**, early uterine contractions push the baby's head against the **cervix**, causing it to stretch and dilate.

- **Stretch-sensitive receptors** in the cervix detect this pressure and send signals to the **hypothalamus**.
- In response, the hypothalamus stimulates the **posterior pituitary** to release **oxytocin**.
- Oxytocin strengthens and increases the frequency of **uterine contractions**, producing further pressure on the cervix and causing additional oxytocin release.

This **self-amplifying cycle** continues until the baby and placenta are expelled, at which point the stimulus is removed, terminating the feedback loop. Positive feedback ensures a **rapid and coordinated completion of childbirth**, unlike negative feedback, which primarily maintains stability.

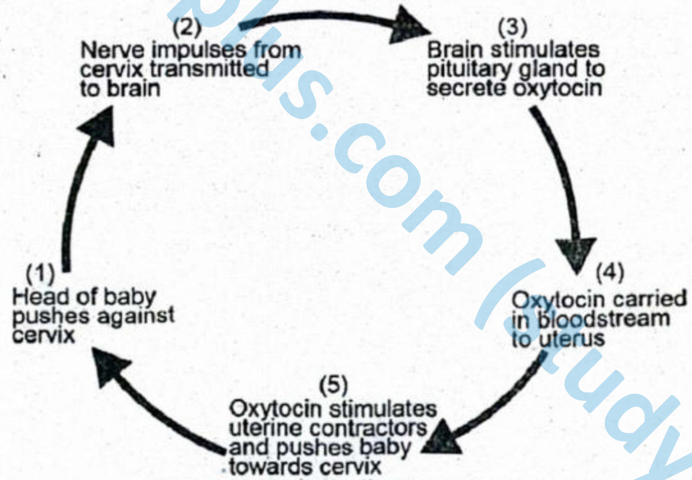



**Figure: Positive feedback mechanism**

### Negative Feedback Mechanism

**Negative feedback** is the most common regulatory mechanism in the body and plays a central role in maintaining stability. In this system, the accumulation of an **end product reverses or counteracts the original stimulus**, restoring the system to its set point.

- **Example: Regulation of Blood Glucose**

The endocrine control of **blood glucose** demonstrates negative feedback:

- When blood glucose levels rise after a meal, **β (beta) cells** in the **islets of Langerhans** of the pancreas detect the increase and secrete **insulin.**
  - Insulin increases **cellular glucose uptake** by activating glucose transporters.
  - It also stimulates enzymes that convert glucose into **glycogen** for storage in liver and muscle cells.
  - These actions collectively lower blood glucose levels to normal.

- Conversely, when blood glucose levels fall, **α (alpha) cells** release **glucagon**, which stimulates **glycogen breakdown** in the liver and the release of glucose into the bloodstream.

Through this tightly regulated negative feedback loop, insulin and glucagon act **antagonistically** to maintain **glucose homeostasis.**

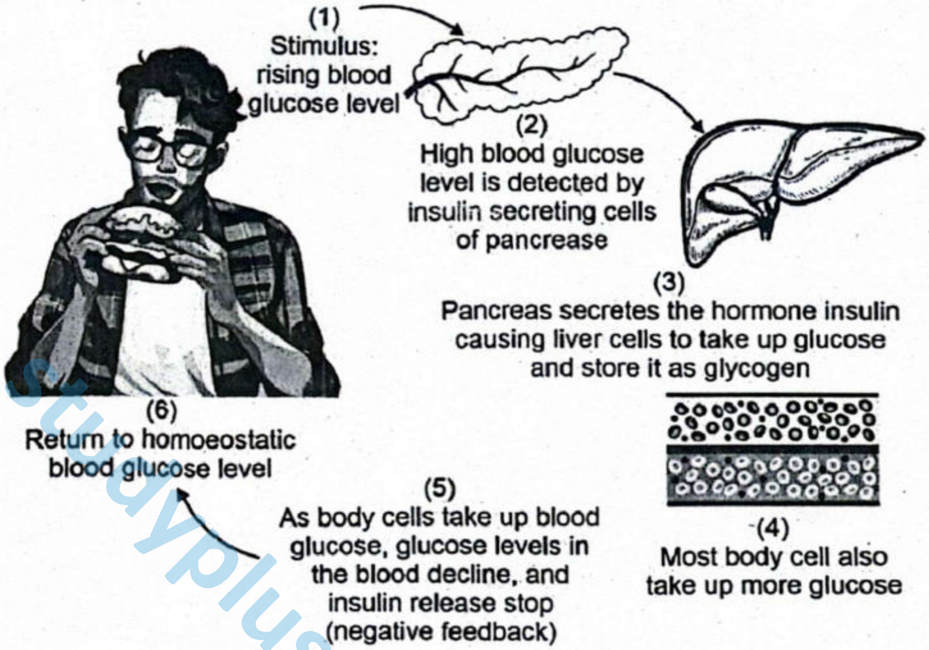



**Figure: Negative feedback mechanism**

## Significance of Feedback Mechanisms

Feedback mechanisms allow the endocrine system to maintain **internal balance** while enabling **amplified responses** when necessary. **Negative feedback** stabilizes physiological processes such as **blood glucose, hormone secretion, and metabolism**, while **positive feedback** facilitates events requiring **progressive amplification,** such as **childbirth.** Together, these mechanisms ensure that hormones act efficiently as **chemical messengers,** keeping the body's internal environment in harmony.

**CRITICAL THINKING**

- Is the sensation of thirst associated with a negative or positive-feedback mechanism?

**Table: Comparison between Positive and Negative Feedback Mechanisms**

| Feature | Positive Feedback Mechanism | Negative Feedback Mechanism |
|---|---|---|
| **Definition** | A mechanism in which the end product amplifies or accelerates the original change. | A mechanism in which the end product reverses or reduces the *original* change. |
| **Effect on Stimulus** | Enhances or intensifies the initial stimulus. | Counteracts or inhibits the initial *stimulus.* |
| **Role in Homeostasis** | Generally does not maintain homeostasis; often used to complete a specific physiological event. | Plays a central role in maintaining homeostasis by stabilizing physiological conditions. |
| **Commonness in the Body** | Relatively rare in healthy individuals. | Very common and found in most regulatory systems of the body. |
| **Typical Outcome** | Leads to a rapid, self-amplifying response until a specific goal is achieved. | Results in the restoration of a system to its normal set point. |
| ***Example*** | *Uterine contractions during* childbirth amplified by oxytocin release. | Regulation of blood glucose levels by insulin and glucagon. |
| **Termination of Response** | Ends only when the triggering event is completed. | Ends automatically once normal conditions are restored. |

## SLOs Based (MCQs)

## HYPOTHALAMUS AS NEUROENDOCRINE CONTROL CENTER AND FEEDBACK MECHANISM

**1. Which of the following best defines a feedback mechanism in biological systems?**
A. A method of nutrient absorption
B. A response that enhances blood circulation
C. A process where the product of a system regulates its own production
D. A mechanism of enzyme denaturation

**2. Which of the following is a characteristic of a positive feedback mechanism?**
A. It restores the original state.
B. It maintains internal stability.
C. It amplifies the initial stimulus.
D. It occurs frequently in all body systems.

**3. Which hormone is released during childbirth as part of the positive feedback mechanism?**
A. Insulin
B. Glucagon
C. Progesterone
D. Oxytocin

**4. What initiates the release of oxytocin during labour?**
A. High levels of glucose
B. Pressure on the cervix
C. Low blood pressure
D. Placental hormone production

**5. Which of the following processes is regulated by a negative feedback mechanism?**
A. Onset of labour
B. Ovulation
C. Blood glucose regulation
D. Lactation

**6. What is the role of insulin in the negative feedback control of blood glucose?**
A. It raises blood glucose by promoting glycogen breakdown.
B. It inhibits glucose transport into cells.
C. It increases blood glucose by triggering gluconeogenesis.
D. It lowers blood glucose by enhancing cellular uptake and glycogen formation.

**7. Which cells release glucagon in response to low blood glucose levels?**
A. Alpha cells of the islets of Langerhans
B. Beta cells of the islets of Langerhans
C. Hepatic cells
D. Neurons in the hypothalamus

**8. How does glucagon help restore blood glucose levels?**
A. By storing glucose as glycogen
B. By increasing insulin secretion
C. By stimulating glucose uptake into muscle cells
D. By activating enzymes that convert glycogen into glucose

**9. Which statement is true about negative feedback?**
A. It amplifies changes and is generally not homeostatic.
B. It is triggered only during emergencies.
C. It reverses deviations to maintain internal balance.
D. It functions independently of endocrine glands.

**10. Which of the following best describes the termination of a positive feedback loop during childbirth?**
A. When oxytocin reaches a threshold level
B. When the placenta begins to produce insulin
C. When the baby and placenta are expelled
D. When cervical dilation stops

**Answer Key**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. C | 2. C | 3. D | 4. B | 5. C | 6. D | 7. A | 8. D | 9. C | 10. C |

## Short Questions

**Q.1 Define a feedback mechanism and describe its role in homeostasis.**

**Ans.** A **feedback mechanism** is a biological process in which the output or product of a system influences its own activity. In homeostasis, feedback mechanisms detect deviations from a physiological set point and trigger responses to restore balance. **Negative feedback** reverses deviations, such as insulin lowering blood glucose when it rises, stabilizing internal conditions. **Positive feedback** amplifies a response until a specific physiological outcome is achieved, like oxytocin-induced uterine contractions during childbirth. Feedback mechanisms are essential for maintaining stable internal environments, regulating temperature, pH, blood pressure, and hormone levels. Without feedback control, the body would be unable to respond effectively to internal or external changes, disrupting homeostasis.

> **MARKING SCHEME (3 Marks)**
> - Definition (2 marks), role in homeostasis (1 mark)

**Q.2 Differentiate between positive and negative feedback mechanisms.**

**Ans.** **Positive feedback** enhances an initial change, creating a self-amplifying loop until a specific event occurs. An example is childbirth: cervical stretching triggers oxytocin release, which intensifies contractions until delivery. **Negative feedback**, in contrast, reverses deviations from a set point to maintain stability. For instance, blood glucose regulation involves insulin lowering glucose when high and glucagon raising it when low. Positive feedback temporarily drives processes toward completion, whereas negative feedback continuously stabilizes internal conditions. Both mechanisms are crucial: positive feedback ensures certain physiological events occur efficiently, and negative feedback maintains homeostasis. Understanding their differences helps explain how the body adapts to changing demands while remaining balanced.

> **MARKING SCHEME (3 Marks)**
> - Definition of positive and negative feedback (1.5 marks each)

**Q.3 How does oxytocin function in the positive feedback loop during childbirth?**

**Ans.** During labor, **stretch receptors** in the cervix detect pressure from the fetus. These receptors send signals to the hypothalamus, stimulating the **release of oxytocin** from the posterior pituitary. Oxytocin increases the **strength and frequency of uterine contractions**, which raises cervical pressure. This further stimulates stretch receptors, creating a **self-amplifying positive feedback loop**. The cycle continues until the baby is delivered, after which the feedback loop ends. This mechanism ensures efficient and timely childbirth by coordinating hormonal and muscular activity, demonstrating the role of positive feedback in amplifying physiological processes to achieve a critical outcome.

> **MARKING SCHEME (3 Marks)**
> - Stretch receptors, oxytocin release, increasing cervical pressure (1 mark each)

**Q.4 Explain how insulin and glucagon maintain blood glucose levels.**

**Ans.** Insulin and glucagon maintain blood glucose through **negative feedback regulation**. When blood glucose rises after a meal, **beta cells** in the pancreas release insulin, promoting glucose uptake by muscle and fat cells and its storage as glycogen in the liver. When blood glucose falls, **alpha cells** release glucagon, stimulating glycogen breakdown into glucose and glucose release from the liver. This **antagonistic hormonal interplay** ensures blood glucose remains within a narrow physiological range, providing a stable energy supply for cells, particularly the brain. Disruption of this balance can lead to hyperglycemia or hypoglycemia, highlighting the importance of insulin and glucagon in homeostasis.

> **MARKING SCHEME (3 Marks)**
> - Role of insulin (1.5 marks), role of glucagon (1.5 marks)

**Q.5 Why is negative feedback considered a stabilizing mechanism?**

**Ans.** Negative feedback is considered stabilizing because it **counteracts deviations from a physiological set point**, restoring internal balance. For example, in thermoregulation, if body temperature rises, mechanisms such as sweating and vasodilation lower it. In blood glucose regulation, insulin lowers elevated glucose, while glucagon raises it when low. By **reducing excessive fluctuations**, negative feedback ensures homeostasis is maintained despite environmental or internal changes. It operates *continuously*, adjusting responses based on feedback signals, which protects the body from extreme conditions and maintains optimal function of vital organs and systems.

> **MARKING SCHEME (3 Marks)**
> - Definition (1 mark), reason (2 marks)

## Exercise

### I Multiple Choice Questions (MCQs)

❖ Select the correct Answer.

1. **Steroid hormones are secreted by**
   A. the adrenal cortex B. the gonads
   C. the thyroid D. both A and B
2. **Examples of posterior pituitary hormones are**
   A. FSH and LH
   B. prolactin and parathormone
   C. melatonin and prostaglandin
   D. ADH and oxytocin
3. **The primary targets for FSH are cells in the**
   A. hypothalamus B. ovary
   C. thyroid D. pituitary
4. **Which of the following controls the activity of all others?**
   A. thyroid B. pituitary
   C. adrenal cortex D. gonads
5. **Which of the following have antagonistic (opposing) effects?**
   A. parathyroid hormone and calcitonin
   B. glucagons and thyroxine
   C. growth hormone and epinephrine
   D. cortisone and ACTH
6. **Which of the following hormones has broadest range of targets?**
   A. ADH B. oxytocin
   C. TSH D. epinephrine
7. **The pancreas increases its output of insulin in response to**
   A. an increase in body temperature
   B. changing cycle of dark and light
   C. a decrease in blood glucose
   D. an increase in blood glucose
8. **Name the gland that is located at the base of the throat, just inferior to the laryngeal prominence (Adam's apple).**
   A. pituitary B. pineal gland
   C. hypothalamus D. thyroid
9. **Action of parathormone in the human body**
   A. decreases blood sodium level
   B. increases blood sodium level
   C. decreases blood calcium level
   D. increases blood calcium level
10. **Which of the following endocrine glands secretes growth hormones?**
    A. thyroid gland B. pituitary gland
    C. adrenal gland D. testes
11. **Which of the following pair of endocrine glands is located in the brain?**
    A. hypothalamus and thymus
    B. pituitary and parathyroid
    C. thyroid and pineal
    D. hypothalamus and pineal
12. **What gland is located just superior to the kidneys?**
    A. pituitary B. adrenal
    C. pancreas D. ovaries
13. **Which if the following gland which can be classified as an endocrine and an exocrine gland?**
    A. thyroid B. thymus
    C. pancreas D. pituitary
14. **Which of these hormones is made by the posterior pituitary?**
    A. FSH B. LH
    C. ACTH D. ADH
15. **Which one of the following is the smallest in the human body?**
    A. pineal gland B. pituitary gland
    C. adrenal gland D. parathyroid glands
16. **Which of the following hormones are responsible for the "fight-or-flight" response?**
    A. epinephrine and norepinephrine
    B. insulin and glucagon
    C. esrtogen and progesterone
    D. thyroxin and melatonin
17. **Calcitonin is a hormone of which of following:**
    A. adrenal cortex B. thyroid gland
    C. pituitary gland D. thymus gland
18. **All of the following are hormones of the anterior pituitary except:**
    A. human growth hormone (GH)
    B. follicle-stimulating hormone (FSH)
    C. parathyroid hormone (PTH)
    D. thyroid-stimulating hormone (TSH)

19. **The secretions from which of these glands differs between males and females?**
A. adrenal
B. parathyroid
C. gonadal
D. pancreas

20. **Calcium level in the blood is regulated by the:**
A. thyroid
B. parathyroid
C. posterior pituitary
D. A and B

**Answer Key**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. D | 2. D | 3. B | 4. B | 5. A | 6. D | 7. D | 8. D | 9. D | 10. B |
| 11. D | 12. B | 13. C | 14. D | 15. A | 16. A | 17. B | 18. C | 19. C | 20. D |

## II Short Answer Questions

**Q.1 Why are hormones called chemical messengers?**

**Ans.** Hormones are called **chemical messengers** because they are secreted by endocrine glands and travel through the bloodstream to distant target organs. They convey information that regulates vital physiological processes such as growth, metabolism, reproduction, water balance, and stress responses. Hormones act at low concentrations and bind to **specific receptors** on target cells, ensuring precise communication. This signaling allows organs and tissues to coordinate their activities without direct nervous control. By facilitating communication between different parts of the body, hormones maintain **homeostasis** and coordinate complex body functions, complementing neural control mechanisms.

**MARKING SCHEME (3 Marks)**
- Definition (2 marks), role in regulation (1 mark)

**Q.2 What are neurosecretory cells?**

**Ans.** **Neurosecretory cells** are specialized neurons that produce and release hormones rather than neurotransmitters. Located mainly in the hypothalamus, they transmit chemical signals directly into the bloodstream through the pituitary stalk. These cells bridge the **nervous and endocrine systems**, translating neural stimuli into hormonal output. For example, they secrete oxytocin and ADH, which act on the kidneys, uterus, and mammary glands. Neurosecretory cells enable the central nervous system to control distant physiological processes via hormonal signals, ensuring integration between neural activity and endocrine regulation.

**MARKING SCHEME (3 Marks)**
- Definition (1.5 marks), function/role (1.5 marks)

**Q.3 Name the hormones of the anterior pituitary gland.**

**Ans.** The anterior pituitary gland secretes several **trophic and non-trophic hormones**, including:

1. **Growth Hormone (GH)** – stimulates growth and protein synthesis.
2. **Thyroid-Stimulating Hormone (TSH)** – regulates thyroid hormone production.
3. **Adrenocorticotropic Hormone (ACTH)** – stimulates adrenal cortex secretion.
4. **Follicle-Stimulating Hormone (FSH)** – controls gamete production.
5. **Luteinizing Hormone (LH)** – triggers ovulation and sex hormone production.
6. **Prolactin (PRL)** – promotes milk synthesis in mammary glands.

These hormones regulate growth, metabolism, reproduction, and other endocrine glands' activities, emphasizing the anterior pituitary's role as a central controller of endocrine function.

**MARKING SCHEME (3 Marks)**
- Name of hormones (1 mark each, total 6 marks)

**Q.4 Why is the anterior lobe of the pituitary gland called the master gland?**

**Ans.** The **anterior pituitary** is termed the master gland because it regulates the function of multiple other endocrine glands through the secretion of **trophic hormones**. These hormones include TSH, ACTH, FSH, LH, and GH, which control the thyroid, adrenal cortex, gonads, and growth processes. By influencing growth, metabolism, reproduction, and stress responses, the anterior pituitary ensures coordination among various endocrine organs. Its ability to **control other glands** makes it central to maintaining homeostasis and regulating complex physiological processes.

**MARKING SCHEME (3 Marks)**
- Definition (1 mark), hormones controlling other glands (2 marks)

**Q.5 Describe the median lobe of the pituitary gland.**

**Ans.** The **median lobe** or intermediate lobe of the pituitary gland is small and relatively underdeveloped in humans. It lies between the anterior and posterior lobes. The main hormone secreted is **melanocyte-stimulating hormone (MSH)**, which regulates **melanin production** in skin melanocytes, affecting pigmentation. Although less prominent than the anterior lobe, the median lobe contributes to the endocrine system by influencing pigmentation and potentially modulating appetite and energy balance. Its function illustrates the diversity of hormonal control within the pituitary gland.

> **MARKING SCHEME (3 Marks)**
> - Description/location (1 mark), hormone produced and function (2 marks)

**Q.6 How is the secretion of ADH controlled?**

**Ans.** The secretion of **antidiuretic hormone (ADH)** is controlled primarily by **osmoreceptors** located in the hypothalamus, which monitor the osmolarity of the blood. When blood osmolarity rises, such as during dehydration, these osmoreceptors detect the increased concentration of solutes and stimulate the **posterior pituitary** to release ADH into the bloodstream. ADH then acts on the **kidney's collecting ducts**, increasing water reabsorption back into the circulation and reducing urine volume. Conversely, when blood osmolarity falls due to excess water, ADH secretion is inhibited, allowing more water to be excreted. This **negative feedback mechanism** maintains blood osmotic balance, controls fluid homeostasis, and prevents excessive dehydration or overhydration.

> **MARKING SCHEME (3 Marks)**
> - Controlling organ (hypothalamus) 1 mark, mechanism 1 mark, role in kidneys 1 mark

**Q.7 Write the differences between:**

**(a) Exocrine and Endocrine glands**

**Ans.** **Exocrine glands** release their secretions through ducts onto external or internal surfaces. Their products are usually enzymes or non-hormonal substances and act locally. Examples include salivary and sweat glands. In contrast, **endocrine glands** secrete hormones directly into the bloodstream without ducts. These hormones act on distant target organs to regulate physiological processes such as growth, metabolism, and reproduction. Examples include the pituitary and thyroid glands. The key distinction lies in **mode of secretion, type of product, and range of action**, with endocrine glands functioning as chemical messengers and exocrine glands performing localized effects.

> **MARKING SCHEME (3 Marks)**
> - Secretion pathway 1 mark, type of product 1 mark, action/target 1 mark

**(b) Steroid hormones and Proteinous hormones**

**Ans.** **Steroid hormones** are lipid-soluble, derived from cholesterol, and can cross cell membranes to bind intracellular receptors. They act directly on DNA in the nucleus to regulate gene expression. Examples include cortisol, estrogen, and testosterone.

**Proteinous hormones** are water-soluble, made of amino acids, and cannot cross membranes. They bind to cell-surface receptors and trigger secondary messenger pathways, like cAMP, to induce cellular responses. Examples include insulin, FSH, and ADH. Steroid hormones produce slower but longer-lasting effects, while proteinous hormones act rapidly but transiently.

> **MARKING SCHEME (3 Marks)**
> - Solubility/source 1 mark, mech[illegible] of action 1 mark, examples 1 mark

**(c) First messenger and Second messenger**

**Ans.** A **first messenger** is the hormone itself that binds to a receptor on the target cell membrane, initiating the signal but unable to enter the cell. Examples include adrenaline and glucagon.

The **second messenger** is an intracellular molecule activated by the hormone-receptor interaction; it transmits and amplifies the signal inside the cell to elicit a response. Examples include cyclic AMP (cAMP) and calcium ions. The first messenger starts the communication from outside the cell, whereas the second messenger ensures the signal spreads and triggers cellular effects efficiently.

> **MARKING SCHEME (3 Marks)**
> - Definition 1 mark, role/mechanism 1 mark, examples 1 mark

**(d) Receptors of *proteinous hormones* and Steroid hormones**

***Ans.*** ***Proteinous* hormone receptors** are located on the cell membrane because these water-soluble hormones cannot cross the membrane. Binding triggers secondary messenger cascades to elicit effects. **Example:** insulin receptor.

> **MARKING SCHEME (3 Marks)**
> - Location 1 mark, *mechanism* 1 mark, example 1 mark

**Steroid hormone receptors** are intracellular, in the cytoplasm or nucleus, since lipid-soluble hormones can enter cells. Binding directly affects gene transcription and protein synthesis. Example: estrogen receptor. The main difference lies in **location, solubility of the hormone, and mechanism of action**, with proteinous hormones acting via indirect pathways and steroid hormones producing direct genomic effects.

**(e) Hypothyroidism and Hyperthyroidism**

**Ans.** **Hypothyroidism** results from under-secretion of thyroid hormones, slowing metabolism. Symptoms include fatigue, weight gain, and cold intolerance. Common causes are iodine deficiency and Hashimoto's disease. Treatment involves hormone replacement therapy.

**MARKING SCHEME (3 Marks)**
- Hormone imbalance 1 mark, symptoms 1 mark, cause/treatment 1 mark

**Hyperthyroidism** is due to excessive thyroid hormone secretion, increasing metabolism. Symptoms include weight loss, anxiety, and heat intolerance. Causes include Graves' disease and thyroid nodules, treated with anti-thyroid drugs or surgery. The key differences lie in **hormone levels, metabolic effects, clinical manifestations, causes, and treatments**, affecting overall physiological activity and requiring opposite management approaches.

**(f) Calcitonin and Parathormone (PTH)**

**Ans.** **Calcitonin** is secreted by the thyroid gland and lowers blood calcium levels by inhibiting bone resorption and promoting calcium storage in bones. It plays a minor role in calcium regulation.

**MARKING SCHEME (3 Marks)**
- Source 1 mark, effect on calcium 1 mark, role in regulation 1 mark

**Parathormone (PTH)** is secreted by the parathyroid glands and increases blood calcium levels by stimulating bone resorption, enhancing calcium reabsorption in the kidneys, and promoting intestinal calcium absorption indirectly. PTH plays a major role in calcium homeostasis. The key difference is that calcitonin reduces calcium levels and PTH raises them, balancing calcium metabolism and maintaining bone and kidney functions.

**(g) Beta and Alpha cells of islets of Langerhans**

**Ans.** **Beta cells** secrete insulin, lowering blood glucose by promoting cellular uptake and glycogen formation, and are more numerous.

**MARKING SCHEME (3 Marks)**
- Secreted hormone 1 mark, effect on glucose 1 mark, function 1 mark

**Alpha cells** secrete glucagon, raising blood glucose by stimulating glycogen breakdown and glucose release, and are fewer in number. Both work antagonistically to maintain glucose homeostasis. Beta cells act after meals to reduce glucose, while alpha cells act during fasting or hypoglycemia to restore blood sugar levels.

**(h) Insulin and Glucagon**

**Ans.** **Insulin** is secreted by pancreatic beta cells after meals, lowering blood glucose by enhancing cellular uptake and glycogen formation.

**MARKING SCHEME (3 Marks)**
- Source 1 mark, *function*/effect 1 mark, timing/condition 1 mark

**Glucagon** is secreted by alpha cells during fasting, raising blood glucose by stimulating glycogenolysis in the liver. Insulin primarily acts to store nutrients, whereas glucagon mobilizes energy reserves. Their antagonistic action maintains glucose homeostasis, preventing hypo- or hyperglycemia.

**(i) Diabetes insipidus and Diabetes mellitus**

**Ans.** **Diabetes insipidus** is caused by ADH deficiency, resulting in excessive urination and thirst without glucose in urine, a disorder of water balance, treated with ADH replacement.

**MARKING SCHEME (3 Marks)**
- Cause 1 mark, main symptom 1 mark, treatment 1 mark

**Diabetes mellitus** is caused by lack of insulin or insulin resistance, leading to high blood glucose and glucose in urine, a disorder of carbohydrate metabolism, managed with insulin or oral hypoglycemics. The main difference is the affected hormone and metabolic pathway.

**(j) Estrogen and Progesterone**

**Ans.** **Estrogen** is secreted mainly by developing ovarian follicles, promoting female *secondary* sexual characteristics and endometrial growth, peaking before ovulation.

**MARKING SCHEME (3 Marks)**
- Source 1 mark, main function 1 mark, role in menstrual cycle/pregnancy 1 mark

**Progesterone** is secreted by the corpus luteum and placenta, preparing and maintaining the endometrium for implantation and supporting pregnancy, peaking after ovulation. Estrogen stimulates growth, while progesterone stabilizes tissue, ensuring proper menstrual cycle regulation and reproductive function.

**(k) Positive and Negative Feedback**

**Ans.** **Positive feedback** amplifies the original stimulus, producing rapid, self-reinforcing effects, e.g., childbirth and blood clotting. It is less common in biological systems.

**Negative feedback** counteracts deviations, maintaining stability and homeostasis, e.g., blood glucose regulation by insulin/glucagon or thyroid hormone regulation. Negative feedback is the predominant mechanism in the body, ensuring physiological parameters remain within optimal ranges. Positive feedback is used for events needing escalation until completion.

> **MARKING SCHEME (3 Marks)**
> - Mechanism 1 mark, effect 1 mark, example 1 mark

**Q.8 What is the relationship between a hormone and a target cell?**

**Ans.** A hormone is a chemical messenger secreted by endocrine glands into the bloodstream. Target cells contain specific receptors that recognize and bind to a hormone. This binding activates intracellular processes, such as enzyme activation, ion channel modulation, or gene expression changes, producing a physiological response. Only cells with the correct receptor for a hormone respond, ensuring specificity. For example, insulin acts only on cells with insulin receptors, promoting glucose uptake, while thyroxine affects cells with thyroid hormone receptors to regulate metabolism. This receptor-hormone interaction underlies precise hormonal regulation throughout the body.

> **MARKING SCHEME (3 Marks)**
> - Hormone definition 1 mark, receptor interaction 1 mark, effect on target cell 1 mark

**Q.9 Locate the endocrine glands in the human body.**

**Ans.** Major endocrine glands are located as follows:

- **Pituitary gland:** base of the brain
- **Hypothalamus:** below the thalamus
- **Thyroid gland:** front of the neck, over the trachea
- **Parathyroid glands:** behind the thyroid
- **Adrenal glands:** atop each kidney
- **Pancreas (Islets of Langerhans):** behind the stomach
- **Gonads:** ovaries in females, testes in males
- **Pineal gland:** near the center of the brain
- **Thymus:** behind the sternum, prominent in children

These glands secrete hormones regulating growth, metabolism, reproduction, and homeostasis..

> **MARKING SCHEME (3 Marks)**
> - Location 1 mark each for 5–6 glands

**Q.10 How does a feedback mechanism regulate the activity of the endocrine system?**

**Ans.** Feedback mechanisms maintain hormone levels within physiological ranges. In **negative feedback**, a rise in hormone concentration inhibits further secretion, e.g., high thyroxine reduces TSH release to prevent overproduction. In **positive feedback**, hormone release is amplified until a specific event occurs, e.g., oxytocin during childbirth increases uterine contractions. These mechanisms ensure hormonal homeostasis, prevent excessive fluctuations, and coordinate complex physiological processes. Negative feedback predominates in maintaining stability, while positive feedback is used for transient, critical events.

> **MARKING SCHEME (3 Marks)**
> - Negative feedback 1 mark, positive feed-back 1 mark, role in homeostasis 1 mark

**Q.11 Make a diagram of positive feedback involving the hormone oxytocin.**

**Ans.** **(Flowchart):**

Head of baby pushes against cervix

↓

Stretch receptors send nerve impulses to hypothalamus

↓

Hypothalamus signals posterior pituitary to release oxytocin

↓

Oxytocin enters bloodstream

↓

Oxytocin stimulates stronger uterine contractions

↓

Increased cervical pressure → more oxytocin released

**Cycle continues until childbirth is complete.**

> **MARKING SCHEME (3 Marks)**
> - 1 mark each for stimulus, nerve impulse, oxytocin release, uterine contraction, feedback loop

**Q.12 Predict the effects that insufficient amounts of FSH and LH will have on the menstrual cycle.**

**Ans.** Low FSH impairs ovarian follicle development and reduces estrogen secretion, preventing proper endometrial growth. Low LH blocks ovulation and corpus luteum formation, reducing progesterone production. These deficiencies disrupt the menstrual cycle, causing irregular or absent periods, poor endometrial preparation, and infertility. Over time, hormonal imbalance may lead to delayed puberty or amenorrhea. Proper levels of FSH and LH are essential for the coordinated cycle of follicular development, ovulation, and luteal phase maintenance.

> **MARKING SCHEME (3 Marks)**
> - Effect of low FSH 1 mark, effect of low LH 1 mark, consequence for menstrual cycle/fertility 1 mark

## III Long Answer Questions

**Q.1 Describe the function of the endocrine system in the human body.**

**Ans.** Please See SLO [B-12-G-25]

**Q.2 State the role of hormone as chemical messenger.**

**Ans.** Please See SLO [B-12-G-23]

**Q.3 Describe chemical nature of hormone and correlate it with important hormones.**

**Ans.** Please See SLO [B-12-G-24]

**Q.4 Describe the hormones secreted by the anterior lobe of pituitary gland.**

**Ans.** Please See SLO [B-12-G-25]

**Q.5 Describe the hormones secreted by the posterior lobe of pituitary gland.**

**Ans.** Please See [B-12-G-25]

**Q.6 Name the hormones, their functions and imbalance of the Pituitary gland.**

**Ans.** Please See SLO [B-12-G-25]

**Q.7 Name the hormones, their functions, and imbalances of the thyroid gland.**

**Ans.** Please See SLO [B-12-G-25]

**Q.8 Name the hormones, their functions, and imbalances of the parathyroid glands.**

**Ans.** Please See SLO [B-12-G-25]

**Q.9 Name the hormones, their functions, and imbalances of the pancreas.**

**Ans.** Please See SLO [B-12-G-25]

**Q.10 Name the hormones, their functions, and imbalances of the adrenal glands.**

**Ans.** Please See SLO [B-12-G-25]

**(a) Name the hormones, their functions, and imbalances of the gonads.**

**Ans.** Please See SLO [B-12-G-25]

**Q.11 Describe the neurosecretory role of hypothalamus.**

**Ans.** Please See SLO [B-12-G-27]

**Q.12 What is feedback mechanism? Describe the positive and negative feedback mechanism with reference to oxytocin, insulin and glucagon.**

**Ans.** Please See SLO [B-12-G-28]